

Project Data Page:





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LP Condensate Flash Vessel

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Dump Condensate Reciever (V8302) -7 percent.pvdb

Date of Analysis : Jun 20,2026 1:37pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2500.000 mm. ID
 Vessel Design Length, Tangent to Tangent 7000.00 mm.
 Distance of Bottom Tangent above Grade 3000.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 0.00 mm.
 Internal Design Temperature 180 °C
 Internal Design Pressure 3.500 bars
 External Design Temperature 100 °C
 External Design Pressure 1.034 bars
 Maximum Allowable Working Pressure 6 bars
 External Max. Allowable Working Pressure 1 bars
 Shop Test Pressure 11.100 bars
 Field Test Pressure 4.550 bars
 Required Minimum Design Metal Temperature -1.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-106 B	...	...	K03006	
Rings	SA-516 70	...	...	K02700	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
skirt-1	...	...	...	1.0000	N/A	No
Skirt-2	...	...	...	1.0000	N/A	No
Bottom Dish	4.021	1.03	6.000	3.0000	No	No
shell 1	3.958	1.03	8.000	3.0000	N/A	No
shell 2	3.676	1.03	8.000	3.0000	N/A	No
shell 3	3.641	1.03	8.000	3.0000	N/A	No
Top dish	3.500	1.03	7.000	3.0000	No	No

Liquid Level: 5550.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.958

**Stiffener Ring Specifications:**

Elevation mm.	Selected Type	Ring Description
5350.00	Bar 111.0 x 14.0	Ring R1\
8250.00	Bar 111.0 x 14.0	Ring R2

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	1700.0	1700.0	8.0	8.0	...	...	0.70	0.70
Skirt	2700.0	1000.0	8.0	8.0	...	...	0.70	0.50
Ellipse	2750.0	50.0	12.0	10.0	7.3	...	0.85	0.85
Cylinder	5750.0	3000.0	12.0	12.0	7.2	...	0.85	0.85
Cylinder	6750.0	1000.0	12.0	12.0	6.9	...	0.85	0.85
Cylinder	9650.0	2900.0	12.0	12.0	6.9	...	0.85	0.85
Ellipse	9700.0	50.0	12.0	10.0	6.7	...	0.85	0.85

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	10	9.52482	1.03421	1.19041
40	Ring	12	11.5439	1.03421	1.18458
Ring	50	12	11.592	1.03421	1.16695
50	60	12	11.592	1.03421	1.16695
60	Ring	12	11.592	1.03421	1.16695
Ring	70	12	9.78276	1.03421	2.09028
70	80	10	9.52482	1.03421	1.19041

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2858.33	3255.76	No Calc	No Calc
Ring	50	2900	3256.04	329.085	434.601
50	60	2900	3256.04	No Calc	No Calc
60	Ring	2900	3256.04	No Calc	No Calc
Ring	70	1658.33	3256.86	260.517	434.601
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2-6"	WRC-107/537	0.828	n/a	Passed
D1-3"	WRC-107/537	0.766	n/a	Passed
N5-2"	WRC-297	0.940	Shell	Passed
N1-16"	WRC-297	0.853	Shell	Passed
N6-6"	WRC-297	0.940	Shell	Passed
N4-3"	WRC-297	0.940	Shell	Passed
N3-20"	WRC-107/537	0.687	n/a	Passed

Wind/Earthquake Shear, Bending:



From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2672.27	9727.8	16632.6	67276.6
20	30	2200	2444.67	9675.58	12283.2	50783.4
30	40	2725	2310.79	9637.9	9905.37	41126.4
40	50	4250	1954.5	9005.14	9798.74	40660.4
50	60	6250	1527.23	6539.09	4576.03	17343.6
60	70	8200	1384.81	5720.59	3119.98	11213.6
70	80	9675	625.88	1642.75	204.417	536.534

Abs Max of the all of the Stress Ratio's : 0.3808

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	2343.0000	mm.
Outside Diameter of Basing	2803.0000	mm.
Nominal Diameter of Bolts	42.0000	mm.
Diameter of Bolt Circle	2663.0000	mm.
Number of Bolts	16	

Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	130.0000	mm.
Height of Gussets	230.0000	mm.
Distance between Gussets	90.0000	mm.
Thickness of Top Plate or Ring	38.0000	mm.
Radial Width of the Top Plate	140.0000	mm.
Number of Gussets	32	

Basing Sketch for Cylindrical Skirts:

Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	2672.	Kgf
Total Earthquake Shear on Support	9728.	Kgf
Wind Moment on Support	24450.	Kg-m.
Earthquake Moment on Support	75094.	Kg-m.
Maximum Initial Bolt Load	3920.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	11946.6	kg.
Shop Test - Fabricated + water (full)	53374.8	kg.
Shipping - Fab. + rem. intls.+ shipping app.	11946.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	26590.0	kg.
Empty - Fab. + intls. + details + wghts.	26590.0	kg.
Operating - Empty + operating liquid (no ca)	51699.6	kg.
Field Test - Empty weight + test medium (full)	65025.2	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-6"	6	OK	...	OK	7.82	OK	Passed
D1-3"	6	...	...	OK	6.40	OK	No Calc[*]
M1-24"	8	OK	...	...	...	OK	Passed
K1-2"	8	...	...	OK	7.80	OK	No Calc[*]
K2-2"	8	...	...	OK	7.80	OK	No Calc[*]
N5-2"	8	...	...	OK	6.42	OK	No Calc[*]
N1-16"	8	OK	...	OK	8.60	OK	Passed
N6-6"	8	OK	...	OK	8.54	OK	Passed
N4-3"	8	...	...	OK	8.55	OK	No Calc[*]
N3-20"	7	OK	...	OK	8.48	OK	Passed
K1A-2"	7	...	...	OK	7.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6 Nozzle : D1-3"
Minimum MAWP Shells/Flanges : 6 Element : Bottom Dish
Minimum MAPnc Shells/Flanges : 9 Element : Top dish

Computed Vessel M.A.W.P. : 6 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N2-6"	0.000	270.000	279.120
30	D1-3"	0.000	90.000	145.262
40	M1-24"	1550.000	90.000	1183.200
40	K1-2"	1600.000	270.000	113.600
40	K2-2"	300.000	270.000	113.600
40	N5-2"	300.000	90.000	93.890
60	N1-16"	4450.001	270.000	760.800
60	N6-6"	5150.000	270.000	284.720
60	N4-3"	5050.001	90.000	164.400
70	N3-20"	0.000	0.000	980.000
70	K1A-2"	0.000	90.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.09	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	219.09	300
N5-2"	2.000 in	XXS	WNF	2.375	11.074	...	...	212.36	300
K1A-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.81	300
D1-3"	3.000 in	160	WNF	3.500	11.125	190.00	10.00	214.44	300
N4-3"	3.000 in	Actual	LWN	4.620	20.574	...	...	213.40	300
N2-6"	6.000 in	160	WNF	6.625	18.237	318.27	10.00	222.99	300
N6-6"	6.000 in	160	WNF	6.625	18.237	318.27	12.00	219.62	300
N1-16"	16.000 in	Actual	WNF	16.000	16.000	812.50	12.00	282.63	150
N3-20"	20.000 in	Actual	WNF	20.000	12.000	622.00	12.00	178.31	150
M1-24"	24.000 in	Actual	WNF	24.000	12.000	909.60	12.00	349.73	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-2"	SA-105	9.000	10.000	...	...	...
K2-2"	SA-105	9.000	10.000	...	...	...
N5-2"	SA-106 B	9.000	10.000	...	...	...
K1A-2"	SA-105	9.000	10.000	...	...	...
D1-3"	SA-106 B	9.000	10.000	8.000	10.000	...
N4-3"	SA-106 B	9.000	10.000	...	...	...
N2-6"	SA-106 B	9.000	10.000	8.000	10.000	...
N6-6"	SA-106 B	12.000	10.000	10.000	12.000	...
N1-16"	SA-516 70	10.000	10.000	12.000	12.000	...
N3-20"	SA-516 70	9.000	10.000	8.000	12.000	...
M1-24"	SA-516 70	9.000	10.000	8.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1-2"	4300.000	270.0	206.38	0.00	shell 1
K2-2"	3000.000	270.0	206.38	0.00	shell 1
N5-2"	3000.000	90.0	200.00	0.00	shell 1
K1A-2"	...	90.0	206.38	0.00	Top dish
D1-3"	...	90.0	200.00	0.00	Bottom Dish
N4-3"	7750.000	90.0	200.02	0.00	shell 3
N2-6"	...	270.0	200.00	0.00	Bottom Dish
N6-6"	7850.001	270.0	204.79	0.00	shell 3
N1-16"	7150.000	270.0	250.00	0.00	shell 3
N3-20"	...	0.0	150.00	0.00	Top dish
M1-24"	4250.000	90.0	300.00	0.00	shell 1

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 8.0mm. THK X 2507.0mm. ID X 1700.0mm.	SA-36
1	SKIRT: 8.0mm. THK X 2507.0mm. ID X 1000.0mm.	SA-516 70
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 2500.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 12.0mm. THK X 2500.0mm. ID X 3000.0mm.	SA-516 70
2	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2700.0mm	...
2	INSULATION: 500mm X 80mm THK	HC
1	INSULATION: 3000mm X 80mm THK	HC
2	BAR RING STIFFENER: 2524.0mm ID X 2746.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 1000mm X 80mm THK	HC
1	LADDER: 3800.0mm	...
1	INSULATION: 2900mm X 80mm THK	HC
1	PLATFORM: 1000mm X 4524mm WIDE	
1	LADDER: 3620.0mm	...
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 2803.0mm. OD X 2343.0 ID X 32.0mm.	SA-36
16	BASERING BOLTS: 42.0mm. DIA	SA-307 A
16	BASERING BOLT NUTS: 42.0mm. DIA	...
1	BASERING TOP RING: 2803.0mm. OD X 2523.0 ID X 38.0mm.	...
32	BASERING GUSSET PLATES: 130.0mm. X 130.0mm. X 230.0mm. X 16.0mm.	...
2	CYLINDRICAL SEGMENT 12.0mm. THK X 3926.9mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 12.0mm. THK X 3926.9mm. X 2900.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Dish	[10]	B	-29	-37	-29	0.850	10.000	0.85	No
Bottom Dish	[7]	B	-23	-42	-29	0.665	12.000	0.85	No
shell 1	[8]	B	-23	-42	-29	0.659	12.000	0.85	No
shell 2	[8]	B	-23	-44	-29	0.630	12.000	0.85	No
shell 3	[8]	B	-23	-44	-29	0.626	12.000	0.85	No
Top dish	[10]	B	-29	-41	-29	0.783	10.000	0.85	No
Top dish	[7]	B	-23	-45	-29	0.612	12.000	0.85	No
Ring R1\	[23]	B	-23	-42	-29	0.659	12.000	0.85	No
Ring R2	[23]	B	-23	-44	-29	0.626	12.000	0.85	No
N2-6"	[1]	B	-29	-42	-29	0.765	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
D1-3"	[1]	B	-29	-42	-29	0.766	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	B	-23	-43	-29	0.644	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1-2"	[1]	A	-3	-29	-29	0.644	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	A	-3	-29	-29	0.656	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-2"	[1]	B	-8	-104	-29	0.025	9.690	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-16"	[1]	B	-23	-48	-29	0.431	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-6"	[1]	B	-23	-44	-29	0.616	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4-3"	[1]	B	-23	-44	-29	0.617	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-20"	[1]	B	-23	-48	-29	0.449	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2"	[1]	A	-8	-29	-29	0.705	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -3 -29

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (bars):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

LP Condensate Flash Vessel

Design Internal Pressure (for Hydrotest) 3.5 bars
 Design Internal Temperature 180.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -1.0 °C
 Type of Construction Welded
 Special Service Air/Water/Steam
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding Y

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	skirt-1
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	2523 mm.
Diameter of Skirt at Base	2523 mm.
Skirt Thickness	8 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	8 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	180 °C
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm ²
Allowable Stress, Operating	114 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	215.4 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	2523 mm.
Diameter of Skirt at Base	2523 mm.
Skirt Thickness	8 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	8 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	180 °C
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	227.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2500 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	180 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	40973 cm ²
Platform Weight	3514.4 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	120 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	2700 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-625 mm.
Height/Length of Liquid	675 mm.
Liquid Density	0.0009565 kg./cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-450 mm.
Height/Length of Insulation	500 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	N2-6"
Dist. from "FROM" Node / Offset dist	300 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	30
Detail Type	Nozzle



Detail ID	D1-3"
Dist. from "FROM" Node / Offset dist	200 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	shell 1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	2500 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	180 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.0009565 kg./cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm ³

Element From Node	40
Detail Type	Ring
Detail ID	Ring R1\
Dist. from "FROM" Node / Offset dist	2600 mm.
Inside Diameter of Ring	2524 mm.
Thickness of Ring	14 mm.
Outside Diameter of Ring	2746 mm.
Material Name	SA-516 70
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	40
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y



Weight of Nozzle (Used if > 0)	397.3	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1-2"	
Dist. from "FROM" Node / Offset dist	1550	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N5-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Weight	
Detail ID	davit	
Dist. from "FROM" Node / Offset dist	1800	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	1500	mm.

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	shell 2	
Distance "FROM" to "TO"	1000	mm.
Inside Diameter	2500	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	3.5	bars
Design Temperature Internal Pressure	180	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-516 70	



Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	375 mm.
Liquid Density	0.0009565 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Insul.0
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1000 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm ³
Element From Node	50
Detail Type	Weight
Detail ID	Lifting Lug
Dist. from "FROM" Node / Offset dist	500 mm.
Miscellaneous Weight	350 Kg
Offset from Element Centerline	0 mm.
Element From Node	50
Detail Type	Weight
Detail ID	Clip Wt
Dist. from "FROM" Node / Offset dist	800 mm.
Miscellaneous Weight	880 Kg
Offset from Element Centerline	0 mm.

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	shell 3
Distance "FROM" to "TO"	2900 mm.
Inside Diameter	2500 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	180 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PLAT 2
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	48532 cm ²
Platform Weight	1753.2 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.



Platform Force Coefficient	1.0	
Ladder Layout Angle	5.0	
Ladder Start Elevation	200	mm.
Ladder End Elevation	4000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1500	mm.
Liquid Density	0.0009565	kg./cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	insulation	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2900	mm.
Thickness of Insulation	80	mm.
Density	0.0003516	kg./cm ³

Element From Node	60	
Detail Type	Ring	
Detail ID	Ring R2	
Dist. from "FROM" Node / Offset dist	1500	mm.
Inside Diameter of Ring	2524	mm.
Thickness of Ring	14	mm.
Outside Diameter of Ring	2746	mm.
Material Name	SA-516 70	
Height of Section Ring	0	mm.
Using Custom Stiffener Section	No	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N1-16"	
Dist. from "FROM" Node / Offset dist	400	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N6-6"	
Dist. from "FROM" Node / Offset dist	1100	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N4-3"	
Dist. from "FROM" Node / Offset dist	1000	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	60
Detail Type	Weight
Detail ID	OVPN116"
Dist. from "FROM" Node / Offset dist	400 mm.
Miscellaneous Weight	2213 Kg
Offset from Element Centerline	2665 mm.
Element From Node	60
Detail Type	Weight
Detail ID	OVPN66"
Dist. from "FROM" Node / Offset dist	1100 mm.
Miscellaneous Weight	780 Kg
Offset from Element Centerline	2506 mm.
Element From Node	60
Detail Type	Weight
Detail ID	Local shell Wt
Dist. from "FROM" Node / Offset dist	400 mm.
Miscellaneous Weight	35 Kg
Offset from Element Centerline	1150 mm.

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2500 mm.
Element Thickness	10 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	3.5 bars
Design Temperature Internal Pressure	180 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Platform
Detail ID	PLAT 3
Dist. from "FROM" Node / Offset dist	800 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	59952 cm ²
Platform Weight	3615 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	4524 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	2262 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	180 mm.
Ladder End Elevation	3800 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	4524 mm.

Element From Node	70
-------------------	----



Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	500 mm.
Thickness of Insulation	80 mm.
Density	0.0003516 kg./cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	N3-20"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	200 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	70
Detail Type	Nozzle
Detail ID	K1A-2"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	70
Detail Type	Weight
Detail ID	Local Shell Wt
Dist. from "FROM" Node / Offset dist	650 mm.
Miscellaneous Weight	35 Kgf
Offset from Element Centerline	0 mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
skirt-1		...	1700	...	1700
Skirt-2		...	2700	...	1000
Bottom Dish		...	2750	...	50
shell 1		...	5750	...	3000
shell 2		...	6750	...	1000
shell 3		...	9650	...	2900
Top dish		...	9700	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
skirt-1		...	8	1	2523	...	1
Skirt-2		...	8	1	2523	...	2
Bottom Dish		4.0212	12	3	2500	117.3	3
shell 1		3.9578	12	3	2500	117.3	4
shell 2		3.6761	12	3	2500	117.3	5
shell 3		3.6409	12	3	2500	117.3	6
Top dish		3.5	12	3	2500	117.3	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
skirt-1		...	No Calc	No Calc	8	No Calc	1
Skirt-2		...	No Calc	No Calc	8	No Calc	2
Bottom Dish		3.5	6.04862	9.37578	10	7.28356	3
shell 1		3.5	7.93079	11.1955	12	7.23662	4
shell 2		3.5	8.21251	11.1955	12	6.93449	5
shell 3		3.5	8.24773	11.1955	12	6.89673	6
Top dish		3.5	6.5698	9.37578	10	6.72821	7

Minimum

6

9

MAWP: 6 bars, limited by: Bottom Dish.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Bottom Dish

Elliptical Head Design Information:			
Design Pressure P	4.021 bars	Int. Design Temperature	180 °C
Material	SA-516 70	Ext. Design Temperature	100 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2500.000 mm.	Actual Stress	84.5 N./mm ²
Spec. Min. Thk t _e , or sa	10.0000 mm.	Spec. Nominal Thk.	12.000 mm.
Surface Area	73020.8 cm ²	SG of Contents	0.958
Element Volume	2291155.2 cm ³	Weight of Contents	2193.7 Kgf
Empty Weight	739.1 Kgf	Operating Weight	2932.8 Kgf
		Comp. All. Ope. Stress	81.7 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.021 \cdot 2506.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 4.021)$$

$$= 4.2836 + 3.0000 = 7.2836 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.521 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (0.997 \cdot 2506.0 + 0.2 \cdot 7.0)$$

$$= 7 - 1 = 6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 2500.0 + 0.2 \cdot 10.0)$$

$$= 9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.021 \cdot (0.997 \cdot 2506.0 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 84.464 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4.021 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.021) + 3.0$$

$$= 7.305 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.462 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1253.0 + 0.6 \cdot 9.0)$$

$$= 8 - 0 = 8 \text{ bars}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2506.0 / (2 \cdot 628.0))^2) / 6$$

$$= 0.996819$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.088 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 7.0$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 7.042$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.665$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.570 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.850	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-37 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.625 mm.

EXTERNAL PRESSURE RESULTS for: Bottom Dish

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	2520.00	360.00	0.0003858	38.57

$$MAEP = B / (K_0 \cdot Do / t) = 38.5716 / (0.9 \cdot 360.0) = 1.1904 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.525	2520.00	386.22	0.0003596	35.95

$$MAEP = B / (K_0 \cdot Do / t) = 35.9532 / (0.9 \cdot 386.2178) = 1.0343 \text{ bars}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: shell 1

Cylindrical Shell Design Information:					
Design Pressure	P	3.958 bars	Int. Design Temperature		180 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2500.000 mm.	Actual Stress		65.1 N./mm ²
Spec. Min. Thk	t _e , or s _a	12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		237881.4 cm ²	SG of Contents		0.958
Element Volume		14728856.0 cm ³	Weight of Contents		14102.1 Kg
Empty Weight		2201.9 Kg	Operating Weight		16304.0 Kg
			Comp. All. Ope. Stress		93.1 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.958 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.958)$$

$$= 4.2366 + 3.0000 = 7.2366 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.458 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1253.0 + 0.6 \cdot 9.0)$$

$$= 8 - 0 = 8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 12.0) / (1250.0 + 0.6 \cdot 12.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.958 \cdot (1253.0 + 0.6 \cdot 9.0)) / (0.85 \cdot 9.0)$$

$$= 65.108 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.478 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 12.0, t_r = 6.974, c = 3.0 mm., E* = 0.85
 Thickness Ratio = t_r * E* / (t_g - c) = 0.659, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.506 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C



Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.659	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-42 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell 1**Cylindrical Shell From 40 to Ring R1\ Ext. Chart: CS-2 at 100 °C**

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	2858.33	280.44	1.1325	0.0002492	24.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9172)/(3*280.4445) = 1.1846 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.544	2524.00	2858.33	295.42	1.1325	0.0002292	22.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.917)/(3*295.4165) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	3255.76	280.44	1.2899	0.0002178	21.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7719)/(3*280.4445) = 1.0351 \text{ bars}$$

Cylindrical Shell From Ring R1\ to 50 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	2900.00	280.44	1.1490	0.0002455	24.55

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.5462)/(3*280.4445) = 1.1669 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.592	2524.00	2900.00	293.76	1.1490	0.0002279	22.79

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7884)/(3*293.7601) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	3256.04	280.44	1.2900	0.0002177	21.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.77)/(3*280.4445) = 1.035 \text{ bars}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: shell 2

Cylindrical Shell Design Information:					
Design Pressure	P	3.676 bars	Int. Design Temperature		180 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		1000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2500.000 mm.	Actual Stress		60.5 N./mm ²
Spec. Min. Thk	t _e , or s _a	12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		79293.8 cm ²	SG of Contents		0.958
Element Volume		4909618.5 cm ³	Weight of Contents		1762.8 Kg
Empty Weight		734.0 Kg	Operating Weight		2496.7 Kg
			Comp. All. Ope. Stress		93.1 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.676 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.676)$$

$$= 3.9345 + 3.0000 = 6.9345 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.176 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1253.0 + 0.6 \cdot 9.0)$$

$$= 8 - 0 = 8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 12.0) / (1250.0 + 0.6 \cdot 12.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.676 \cdot (1253.0 + 0.6 \cdot 9.0)) / (0.85 \cdot 9.0)$$

$$= 60.474 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.478 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 12.0, t_r = 6.671, c = 3.0 mm., E* = 0.85
 Thickness Ratio = t_r * E* / (t_g - c) = 0.63, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.225 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C



Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.630	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell 2**Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 100 °C**

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	2900.00	280.44	1.1490	0.0002455	24.55

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.5462)/(3*280.4445) = 1.1669 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.592	2524.00	2900.00	293.76	1.1490	0.0002279	22.79

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7884)/(3*293.7601) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	3256.04	280.44	1.2900	0.0002177	21.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.77)/(3*280.4445) = 1.035 \text{ bars}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell 3

Cylindrical Shell Design Information:					
Design Pressure	P	3.641 bars	Int. Design Temperature		180 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		2900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2500.000 mm.	Actual Stress		59.9 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		229952.0 cm ²	SG of Contents		0.958
Element Volume		14237894.0 cm ³	Weight of Contents		7051.1 Kg
Empty Weight		2128.5 Kg	Operating Weight		9179.6 Kg
			Comp. All. Ope. Stress		93.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3.641 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.641)$$

$$= 3.8967 + 3.0000 = 6.8967 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.141 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1253.0 + 0.6 \cdot 9.0)$$

$$= 8 - 0 = 8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 12.0) / (1250.0 + 0.6 \cdot 12.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (3.641 \cdot (1253.0 + 0.6 \cdot 9.0)) / (0.85 \cdot 9.0)$$

$$= 59.894 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.478 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 12.0, t_r = 6.633, c = 3.0 mm., E* = 0.85
 Thickness Ratio = t_r * E* / (t_g - c) = 0.626, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.189 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C



Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.626	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: shell 3**Cylindrical Shell From 60 to Ring R2 Ext. Chart: CS-2 at 100 °C**

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	2900.00	280.44	1.1490	0.0002455	24.55

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.5462)/(3*280.4445) = 1.1669 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.592	2524.00	2900.00	293.76	1.1490	0.0002279	22.79

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7884)/(3*293.7602) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	3256.04	280.44	1.2900	0.0002177	21.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.77)/(3*280.4445) = 1.035 \text{ bars}$$

Cylindrical Shell From Ring R2 to 70 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	1658.33	280.44	0.6570	0.0004398	43.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9682)/(3*280.4445) = 2.0903 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.783	2524.00	1658.33	372.12	0.6570	0.0002887	28.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.8668)/(3*372.12) = 1.0343 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	2524.00	3256.86	280.44	1.2904	0.0002177	21.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.7643)/(3*280.4445) = 1.0347 \text{ bars}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top dish

Elliptical Head Design Information:					
Design Pressure	P	3.500 bars	Int. Design Temperature		180 °C
Material		SA-516 70	Ext. Design Temperature		100 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2500.000 mm.	Actual Stress		73.5 N./mm ²
Spec. Min. Thk t _e , or s _a		10.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		73020.8 cm ²	SG of Contents		0.000
Element Volume		2291155.2 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		739.1 Kg _f	Operating Weight		739.1 Kg _f
			Comp. All. Ope. Stress		81.7 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3.5 \cdot 2506.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 3.5)$$

$$= 3.7282 + 3.0000 = 6.7282 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (0.997 \cdot 2506.0 + 0.2 \cdot 7.0)$$

$$= 7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 2500.0 + 0.2 \cdot 10.0)$$

$$= 9 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3.5 \cdot (0.997 \cdot 2506.0 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 73.517 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (3.5 \cdot 1253.0) / (138.0 \cdot 0.85 - 0.6 \cdot 3.5) + 3.0$$

$$= 6.746 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 9.0) / (1253.0 + 0.6 \cdot 9.0)$$

$$= 8 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2506.0 / (2 \cdot 628.0))^2) / 6$$

= 0.996819

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.088 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.0$, $t_r = 6.444$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.783$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 6.482$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.612$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -45 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.049 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.783	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.625 mm.

EXTERNAL PRESSURE RESULTS for: Top dish

Elliptical Head From 70 to 80 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	2520.00	360.00	0.0003858	38.57

$$MAEP = B/(K0 \cdot Do/t) = 38.5716 / (0.9 \cdot 360.0) = 1.1904 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.525	2520.00	386.22	0.0003596	35.95

$$MAEP = B/(K0 \cdot Do/t) = 35.9532 / (0.9 \cdot 386.2178) = 1.0343 \text{ bars}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.863 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11.845 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.653 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	6.049 bars

User Defined Hydrostatic Test Pressure at High Point 11.100 bars

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	167.0	234.0	0.714	11.35
shell 1	139.8	234.0	0.598	11.35
shell 2	139.8	234.0	0.598	11.35
shell 3	139.8	234.0	0.598	11.35
Top dish	167.0	234.0	0.714	11.35

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-6"	Nozzle	118.00	118.00	1.000
N2-6"	Pad	138.00	138.00	1.000
D1-3"	Nozzle	118.00	118.00	1.000
D1-3"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N5-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
N1-16"	Pad	138.00	138.00	1.000
N6-6"	Nozzle	118.00	118.00	1.000
N6-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N3-20"	Nozzle	138.00	138.00	1.000
N3-20"	Pad	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1\	138.00	138.00	1.000
Ring R2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
shell 1	138.00	138.00	1.000

shell 2	138.00	138.00	1.000
shell 3	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-6"	5.53	216.00	0.026
D1-3"	4.73	216.00	0.022
M1-24"	28.36	234.00	0.121
K1-2"	2.40	225.00	0.011
K2-2"	2.40	225.00	0.011
N5-2"	3.08	216.00	0.014
N1-16"	13.96	234.00	0.060
N6-6"	5.53	216.00	0.026
N4-3"	2.78	216.00	0.013
N3-20"	23.56	234.00	0.101
K1A-2"	2.40	225.00	0.011

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.863 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	4.550 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	12.073 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.653 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	5.005 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5.005 bars
App 27-4	= M.A.W.P.	6.049 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 3.5 • 1.0

= 4.550 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	113.6	234.0	0.485	5.41
shell 1	87.9	234.0	0.376	5.34
shell 2	83.0	234.0	0.355	5.05
shell 3	81.4	234.0	0.348	4.95
Top dish	98.0	234.0	0.419	4.67

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-6"	Nozzle	118.00	118.00	1.000
N2-6"	Pad	138.00	138.00	1.000
D1-3"	Nozzle	118.00	118.00	1.000
D1-3"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
N5-2"	Nozzle	118.00	118.00	1.000
N1-16"	Nozzle	138.00	138.00	1.000
N1-16"	Pad	138.00	138.00	1.000
N6-6"	Nozzle	118.00	118.00	1.000
N6-6"	Pad	138.00	138.00	1.000
N4-3"	Nozzle	118.00	118.00	1.000
N3-20"	Nozzle	138.00	138.00	1.000
N3-20"	Pad	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1\	138.00	138.00	1.000
Ring R2	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



Bottom Dish	138.00	138.00	1.000
shell 1	138.00	138.00	1.000
shell 2	138.00	138.00	1.000
shell 3	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-6"	2.64	216.00	0.012
D1-3"	2.25	216.00	0.010
M1-24"	13.35	234.00	0.057
K1-2"	1.13	225.00	0.005
K2-2"	1.13	225.00	0.005
N5-2"	1.45	216.00	0.007
N1-16"	6.09	234.00	0.026
N6-6"	2.41	216.00	0.011
N4-3"	1.21	216.00	0.006
N3-20"	9.69	234.00	0.041
K1A-2"	0.99	225.00	0.004

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

MDMT Calculations for: Ring R1\, Shell to Ring Junction, Curve: B

Govern. thk, $t_g = 12.0$, $t_r = 6.974$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.659$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1\, SA-516 70 , Bar Ring: 111.0 x 14.0 mm.

Material treatment options:

Effective Length of Shell: 166 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	14.921	4.5000	67.145
Ring :	15.540	64.5000	1002.330
Total:	30.461		1069.475

Centroid of Ring plus Shell: 35 mm.

	Inertia	Distance	A*Dist ²
Shell:	1.007	30.6095	13980.228
Ring :	159.557	-29.3905	13423.463
Total:	160.564		27403.691

Available Moment of Inertia, Ring + Shell: 0.435E+03 cm⁴

Required Stress in Ring plus Shell Breq 20.52 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002050

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (2524.0002^2 \cdot 2879.1667(9.0 + 15.54/2879.1667)0.000205) / 10.9$$

$$= 329 \text{ cm}^4$$

MDMT Calculations for: Ring R2, Shell to Ring Junction, Curve: B

Govern. thk, $t_g = 12.0$, $t_r = 6.633$, $c = 3.0$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.626$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R2, SA-516 70 , Bar Ring: 111.0 x 14.0 mm.

Material treatment options:

Effective Length of Shell: 166 mm.

	Area (cm ²)	Distance (mm.)	Area*Dist
Shell:	14.921	4.5000	67.145
Ring :	15.540	64.5000	1002.330
Total:	30.461		1069.475

Centroid of Ring plus Shell: 35 mm.



	Inertia	Distance	A*Dist ²
Shell:	1.007	30.6095	13980.228
Ring :	159.557	-29.3905	13423.463
Total:	160.564		27403.691

Available Moment of Inertia, Ring + Shell: 0.435E+03 cm⁴

Required Stress in Ring plus Shell Breq 20.22 N./mm²
 Required Strain in Ring plus Shell Areq 0.0002020

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (2524.0002^2 \cdot 2279.1665(9.0 + 15.54/2279.1665)0.000202) / 10.9$$

$$= 261 \text{ cm}^4$$

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	2520	7	0.0003858	38.5716
40	Ring	2858.33	2524	9	0.00024923	24.9172
Ring	50	2900	2524	9	0.00024552	24.5462
50	60	2900	2524	9	0.00024552	24.5462
60	Ring	2900	2524	9	0.00024552	24.5462
Ring	70	1658.33	2524	9	0.00043978	43.9682
70	80	No Calc	2520	7	0.0003858	38.5716

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	10	9.52482	1.03421	1.19041
40	Ring	12	11.5439	1.03421	1.18458
Ring	50	12	11.592	1.03421	1.16695
50	60	12	11.592	1.03421	1.16695
60	Ring	12	11.592	1.03421	1.16695
Ring	70	12	9.78276	1.03421	2.09028
70	80	10	9.52482	1.03421	1.19041

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2858.33	3255.76	No Calc	No Calc
Ring	50	2900	3256.04	329.085	434.601
50	60	2900	3256.04	No Calc	No Calc
60	Ring	2900	3256.04	No Calc	No Calc
Ring	70	1658.33	3256.86	260.517	434.601
70	80	No Calc	No Calc	No Calc	No Calc

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : Dump Condensate Reciever (V8302) -7 percent

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External Pressure Calculations: Step: 11 1:37pm Jun 20,2026



HEXAGON

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	1757.54	...	1653.71	...	123.027
20	30	489.902	...	428.831	...	34.2931
30	40	739.109	2291155	554.332	2307099	51.7377
40	50	2201.92	14728856	1653.41	14799639	154.135
50	60	733.975	4909618	551.139	4933213	51.3783
60	70	2128.53	14237894	1598.3	14306318	148.997
70	80	739.109	2291155	554.332	2307099	51.7377
Total		8790	38458680.00	6994	38653372.00	615

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	3760.38	...	...	0.19555E-04	PLAT:[1 OF 1]
30	Liqd	2193.66	...	-312.5	...	Liquid 30
30	Insl	278.178	...	-200	...	Ins: 10
30	Nozl	41.5839	-300	-751.733	...	N2-6"
30	Nozl	42.8293	200	-1459.89	...	D1-3"
40	Liqd	14102.1	...	1500	...	Liquid 40
40	Insl	691.125	...	1500	...	Ins: 20
40	Ring	106.683	...	2600	...	Ring R1\
40	Nozl	425.113	1400	1500	...	M1-24"
40	Nozl	11.0224	-1353.19	1550	...	K1-2"
40	Nozl	11.0224	-1353.19	250	...	K2-2"
40	Nozl	7.22035	1350	250	...	N5-2"
40	Wght	50	1500	1800	...	davit
50	Liqd	1762.77	...	187.5	...	Liquid 50
50	Insl	230.375	...	500	...	Insul.
50	Wght	350	...	500	...	Lifting Lug
50	Wght	880	...	800	...	Clip Wt
60	Plat	1875.98	-1332.92	...	...	PLAT 2
60	Liqd	7051.06	...	750	...	Liquid: 60
60	Insl	668.088	...	1450	...	insulation
60	Ring	106.683	...	1500	...	Ring R2
60	Nozl	151.436	-1375.01	400	...	N1-16"
60	Nozl	42.9637	-1352.39	1100	...	N6-6"
60	Nozl	19.7248	1350.01	1000	...	N4-3"
60	Wght	2213	-2665	400	...	OVP_N1_16"
60	Wght	780	-2506	1100	...	OVP_N6_6"
60	Wght	35	-1150	400	...	Local shell Wt
70	Plat	3868.1	...	800	...	PLAT 3
70	Insl	278.178	...	250	...	Ins: 30
70	Nozl	214	...	745.006	...	N3-20"
70	Nozl	10.9634	600	696.48	...	K1A-2"
70	Wght	35	...	650	...	Local Shell Wt

Total Weight of Each Detail Type:

Platforms	9504.5
Liquid	25109.6
Insulation	2145.9
Stiffeners	213.4
Nozzles	977.9
Weights	4343.0

Sum of the Detail Weights 42294.2 kg.

Weight Summation Results: (kg.)



	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9405.4	9405.4	9405.4	9405.4	9405.4	9405.4
Stif. Rings	213.4	213.4	213.4	213.4	213.4	213.4
Nozzles	977.9	977.9	977.9	977.9	977.9	977.9
Wld Weights	1350.0	1350.0	1350.0	1350.0	1350.0	1350.0
Platforms	...	...	...	9504.5	9504.5	9504.5
Insulation	...	...	...	2145.9	2145.9	2145.9
Empty Weights	...	...	...	2993.0	2993.0	...
Test Weights	...	2993.0	...	...	...	...
Ope Weights	...	...	...	...	...	2993.0
Ope. Liquid	...	...	...	...	...	25109.6
Test Liquid	...	38435.2	...	...	...	...
Totals	11946.6	53374.8	11946.6	26590.0	26590.0	51699.6

Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	11946.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	53374.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	11946.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	26590.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	26590.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	51699.6 kg.
Field Test Wt.	- Empty weight + test medium (full)	65025.2 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	20765.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	32228.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	14535.3 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	134746	133892
20	30	79262.4	78759.7
30	40	73020.8	71676.1
40	50	237881	235619
50	60	79293.8	78539.8
60	70	229952	227765
70	80	73020.8	71676.1

Total	907177.3	897928.2 cm ²
Total	976.5	966.5 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	1880.56	1880.56	1880.56	...	1880.56
20	30	524.195	524.195	524.195	...	524.195
30	40	875.26	7107.48	7203.57	3.9094	4913.82



40	50	2967.12	17760.4	18378.1	650.088	3658.24
50	60	2015.35	4008.49	7152.35	...	2245.73
60	70	5626.33	15221.5	22399.6	10633	8170.4
70	80	1050.81	5197.09	7486.85	6.57816	5197.09

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	26590	51699.6	65025.2
20	30	24709.5	49819.1	63144.7
30	40	24185.3	49294.9	62620.5
40	50	19271.5	42187.4	55416.9
50	60	15613.2	24427	37038.8
60	70	13367.5	20418.5	29886.4
70	80	5197.09	5197.09	7486.85

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	11293.6	11293.6	11293.6
20	30	11293.6	11293.6	11293.6
30	40	11293.6	11293.6	11293.6
40	50	11289.7	11289.7	11289.7
50	60	10639.6	10639.6	10639.6
60	70	10639.6	10639.6	10639.6
70	80	6.57816	6.57816	6.57816

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2-6"	44.32	51.00	180	300	GR 1.1	...	...	...	...
D1-3"	44.32	51.00	180	300	GR 1.1	...	...	...	...
M1-24"	14.60	19.60	180	150	GR 1.1	...	...	...	...
K1-2"	44.32	51.00	180	300	GR 1.1	...	...	...	...
K2-2"	44.32	51.00	180	300	GR 1.1	...	...	...	...
N5-2"	44.32	51.00	180	300	GR 1.1	...	...	...	...
N1-16"	14.60	19.60	180	150	GR 1.1	...	...	...	...
N6-6"	44.32	51.00	180	300	GR 1.1	...	...	...	...
N4-3"	44.32	51.00	180	300	GR 1.1	...	...	...	...
N3-20"	14.60	19.60	180	150	GR 1.1	...	...	...	...
K1A-2"	44.32	51.00	180	300	GR 1.1	...	...	...	...
Min Rating	14.600	19.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 7.60774 Hz.

The Natural Frequency for the Vessel (Ope...) is 6.01115 Hz.

The Natural Frequency for the Vessel (Filled) is 5.13988 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.219 \cdot 0.938) / (1 + 1.7 \cdot 3.4 \cdot 0.219)))$
 $= \min(0.85, 0.893)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	6.011 Hz
Natural Frequency of Vessel (Empty)	7.608 Hz
Natural Frequency of Vessel (Test)	5.140 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.550
Structure Height to Diameter ratio	4.022
Height to top of Structure	10415.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850.0 + 300.0 = 1150.0$ mm.
 $= 3.773$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.773 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

$= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16.0 \cdot 0.85 \cdot 0.55 \cdot 60.941$

$= 456.1 \text{ lbs. } [206.9] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
skirt-1	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-2	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Dish	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
shell 1	4550.0	0.000	0.000	0.000	0.849	1.000	78.120
shell 2	6550.0	0.000	0.000	0.000	0.916	1.000	78.120
shell 3	8500.0	0.000	0.000	0.000	0.967	1.000	78.120
Top dish	10276.6	0.000	0.000	0.000	1.007	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/9700.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[8.66/31.82^2] \cdot 10^4 = 85.494$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$

$= 49778 / (9700.0 \cdot 2630.625^2)$

$= 0.74156\text{E-}06$

Since Vp is $> 0.400\text{E-}06$ no further vibration analysis is required !

**Platform Load Calculations**

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	40972.80	3000.00	78.12	320.08	1.00
PLAT 2	48531.80	7050.00	78.12	379.13	1.00
PLAT 3	59951.68	10750.00	78.12	468.34	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
davit	0.00	4850.00	78.12	0.00
Lifting Lug	0.00	6550.00	78.12	0.00
Clip Wt	0.00	6850.00	78.12	0.00
OVPN116"	0.00	7450.00	78.12	0.00
OVPN66"	0.00	8150.00	78.12	0.00
Local shell W	0.00	7450.00	78.12	0.00
Local Shell W	0.00	10600.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 6.01115 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1150	3330.36	56616.1	78.12	227.596
20	30	2500	3330.36	33303.6	78.12	133.88
30	40	3025	3537.6	1768.8	78.12	356.289
40	50	4550	3542.88	106286	78.12	427.27
50	60	6550	3542.88	35428.8	78.12	142.423
60	70	8500	3542.88	102744	78.12	758.929
70	80	10276.6	3537.6	21634.5	78.12	625.88

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(33.87470.75)$
 $= 0.281$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/6.011)$
 $= 0.166$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.281, 0.166)$ per 12.8.2
 $= 0.166$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 2.5 \cdot 0.157 \cdot 51700(1 + 2 \cdot 0.1) / (1.25/1.25) \\
 &= 9727.805 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 51700 = 16213.01 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 51700$
 $= 3039.939 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 9727.805 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 51700$
 $= 1621.3 \text{ Kgf}$

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 6.01115 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	1880.56	52.2236	51.8595	1621.3
20	30	2200	524.195	37.6769	37.4142	1562.33
30	40	2725	7107.48	632.765	77.3793	1545.89
40	50	4250	17760.4	2466.04	409.114	1323
50	60	6250	4008.49	818.505	408.651	766.032
60	70	8200	15221.5	4077.84	1496.79	640.326
70	80	9675	5197.09	1642.75	329.835	162.981



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2672.27	9727.8	16632.6	67276.6
20	30	2200	2444.67	9675.58	12283.2	50783.4
30	40	2725	2310.79	9637.9	9905.37	41126.4
40	50	4250	1954.5	9005.14	9798.74	40660.4
50	60	6250	1527.23	6539.09	4576.03	17343.6
60	70	8200	1384.81	5720.59	3119.98	11213.6
70	80	9675	625.88	1642.75	204.417	536.534

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	2672.27	0.0062002	0.023656	0.26499E-04
20	30	2444.67	0.03851	0.056358	0.38475E-04
30	40	2310.79	0.057323	0.058296	0.39047E-04
40	50	1954.5	0.12584	0.20768	0.58186E-04
50	60	1527.23	0.23726	0.2677	0.61642E-04
60	70	1384.81	0.35944	0.45391	0.65506E-04
70	80	625.88	0.45554	0.45718	0.65506E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	359.377	2246.1
20	30	359.377	2246.1
30	40	381.74	2385.87
40	50	382.31	2389.44
50	60	382.31	2389.44
60	70	382.31	2389.44
70	80	381.74	2385.87

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 48.500 Actual deflection : 0.457 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	632.088	553.291	39616820	34705824
20	30	632.088	553.291	39616820	34705824
30	40	788.539	552.638	49285176	34623124
40	50	947.004	711.103	59190564	44551748
50	60	947.004	711.103	59190564	44551748
60	70	947.004	711.103	59190564	44551748
70	80	788.539	552.638	49285176	34623124

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-81.6094	-83.2155
20	30	82.7985	140.397	-81.6094	-83.2155
30	40	140.757	238.676	-81.7066	-83.3146
40	50	140.757	238.676	-93.079	-98.0261
50	60	140.757	238.676	-93.079	-98.0261
60	70	140.757	238.676	-93.079	-98.0261
70	80	140.757	238.676	-81.7066	-83.3146

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	31.2568	-9.33436	40.6339
40	50	24.2953	-7.2773	31.5839
50	60	24.2953	-7.2773	31.5839
60	70	24.2953	-7.2773	31.5839
70	80	31.2568	-9.33436	40.6339

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.71296	-9.16353	-11.5254	3.19117	3.19117
20	30	-4.37964	-8.8302	-11.1921	3.19117	3.19117
30	40	-4.2918	-4.2918	-4.2918	3.19879	3.19879
40	50	-2.65773	-2.65773	-2.65773	2.48506	2.48506
50	60	-2.15322	-2.15322	-2.15322	2.34197	2.34197
60	70	-1.84351	-1.84351	-1.84351	2.34197	2.34197
70	80	-0.92225	-0.92225	-0.92225	0.0018632	0.0018632

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	3.19117	4.69979	19.01	2.3499	...
20	30	3.19117	3.47078	14.3496	1.73539	...
30	40	3.19879	2.80559	11.6486	1.4028	...
40	50	2.48506	2.15688	8.95007	1.07844	...
50	60	2.34197	1.00727	3.81763	0.50363	...
60	70	2.34197	0.68676	2.46831	0.34338	...
70	80	0.0018632	0.057899	0.15197	0.02895	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	5.70632
20	30	...	...	...	4.36844
30	40	...	...	...	3.60265
40	50	...	...	...	2.77024
50	60	...	...	...	1.15906
60	70	...	...	...	0.712
70	80	...	...	...	0.030512

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.28737
20	30	...	0.27692
30	40	...	0.27433
40	50	...	0.18245
50	60	...	0.10564
60	70	...	0.088307
70	80	...	0.028922

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.18	95.76	-12.60	81.61	0.0332	0.1544
20	2.28	82.80	-11.04	81.61	0.0276	0.1353
30	1.71	140.76	-10.30	81.71	0.0122	0.1260
40	1.98	140.76	-7.30	93.08	0.0141	0.0784
50	1.20	140.76	-5.50	93.08	0.0085	0.0591
60	1.19	140.76	-4.87	93.08	0.0084	0.0523
70		140.76	-0.98	81.71		0.0120

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.47	95.76	-13.32	81.61	0.0467	0.1633
20	3.46	82.80	-11.66	81.61	0.0418	0.1429
30	2.78	140.76	-10.82	81.71	0.0198	0.1324
40	2.78	140.76	-7.73	93.08	0.0198	0.0831
50	1.45	140.76	-5.55	93.08	0.0103	0.0596
60	1.30	140.76	-4.81	93.08	0.0092	0.0517
70		140.76	-0.93	81.71		0.0113

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------



10		95.76	-17.05	81.61		0.2090
20		82.80	-15.49	81.61		0.1898
30	1.71	140.76	-10.30	81.71	0.0122	0.1260
40	1.98	140.76	-7.30	93.08	0.0141	0.0784
50	1.20	140.76	-5.50	93.08	0.0085	0.0591
60	1.19	140.76	-4.87	93.08	0.0084	0.0523
70		140.76	-0.98	81.71		0.0120

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.32	95.76	-31.08	81.61	0.1392	0.3808
20	8.99	82.80	-26.09	81.61	0.1085	0.3197
30	10.83	140.76	-18.86	81.71	0.0769	0.2309
40	8.96	140.76	-13.91	93.08	0.0637	0.1494
50	4.11	140.76	-8.21	93.08	0.0292	0.0882
60	3.06	140.76	-6.57	93.08	0.0217	0.0705
70		140.76	-1.05	81.71		0.0128

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-17.07	83.22		0.2051
20		140.40	-16.12	83.22		0.1937
30	0.31	238.68	-8.89	83.31	0.0013	0.1067
40	0.91	238.68	-6.22	98.03	0.0038	0.0635
50	0.69	238.68	-5.00	98.03	0.0029	0.0510
60	0.84	238.68	-4.53	98.03	0.0035	0.0462
70		238.68	-0.95	83.31		0.0114

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-14.72	83.22		0.1768
20		140.40	-14.38	83.22		0.1728
30		238.68	-7.49	83.31		0.0899
40		238.68	-5.14	98.03		0.0525
50	0.19	238.68	-4.50	98.03	0.0008	0.0459
60	0.50	238.68	-4.19	98.03	0.0021	0.0427
70		238.68	-0.92	83.31		0.0111

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-17.05	81.61		0.2090
20		82.80	-15.49	81.61		0.1898
30	32.97	140.76		81.71	0.2342	
40	26.28	140.76		93.08	0.1867	
50	25.49	140.76		93.08	0.1811	
60	25.48	140.76		93.08	0.1810	
70	30.39	140.76		81.71	0.2159	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.32	95.76	-31.08	81.61	0.1392	0.3808
20	8.99	82.80	-26.09	81.61	0.1085	0.3197
30	42.09	140.76		81.71	0.2990	
40	33.26	140.76		93.08	0.2363	
50	28.41	140.76		93.08	0.2018	
60	27.35	140.76		93.08	0.1943	
70	30.52	140.76		81.71	0.2168	



Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-17.05	83.22		0.2049
20		82.80	-15.49	83.22		0.1862
30		140.76	-19.63	83.31		0.2356
40		140.76	-14.58	98.03		0.1487
50		140.76	-12.78	98.03		0.1304
60		140.76	-12.15	98.03		0.1239
70		140.76	-10.32	83.31		0.1238

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.32	95.76	-31.08	83.22	0.1392	0.3735
20	8.99	82.80	-26.09	83.22	0.1085	0.3136
30	1.50	140.76	-28.20	83.31	0.0106	0.3385
40	1.68	140.76	-21.19	98.03	0.0120	0.2161
50		140.76	-15.48	98.03		0.1580
60		140.76	-13.84	98.03		0.1412
70		140.76	-10.38	83.31		0.1246

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-17.07	83.22		0.2051
20		140.40	-16.12	83.22		0.1937
30	40.94	238.68		83.31	0.1715	
40	32.49	238.68		98.03	0.1361	
50	32.28	238.68		98.03	0.1352	
60	32.43	238.68		98.03	0.1359	
70	39.74	238.68		83.31	0.1665	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-14.72	83.22		0.1768
20		140.40	-14.38	83.22		0.1728
30	39.54	238.68		83.31	0.1657	
40	31.41	238.68		98.03	0.1316	
50	31.77	238.68		98.03	0.1331	
60	32.08	238.68		98.03	0.1344	
70	39.71	238.68		83.31	0.1664	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.90	81.61		0.0969
20		82.80	-7.57	81.61		0.0928
30	30.16	140.76		81.71	0.2143	
40	24.12	140.76		93.08	0.1714	
50	24.48	140.76		93.08	0.1739	
60	24.79	140.76		93.08	0.1761	
70	30.34	140.76		81.71	0.2155	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-8.02	81.61		0.0983
20		82.80	-7.73	81.61		0.0947
30	28.25	140.76		81.71	0.2007	
40	22.30	140.76		93.08	0.1584	
50	22.68	140.76		93.08	0.1611	



60	22.91	140.76	93.08	0.1628
70	30.61	140.76	81.71	0.2175

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-12.35	81.61		0.1514
20		82.80	-12.02	81.61		0.1473
30	30.16	140.76		81.71	0.2143	
40	24.12	140.76		93.08	0.1714	
50	24.48	140.76		93.08	0.1739	
60	24.79	140.76		93.08	0.1761	
70	30.34	140.76		81.71	0.2155	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.90	81.61		0.0969
20		82.80	-7.57	81.61		0.0928
30	30.16	140.76		81.71	0.2143	
40	24.12	140.76		93.08	0.1714	
50	24.48	140.76		93.08	0.1739	
60	24.79	140.76		93.08	0.1761	
70	30.34	140.76		81.71	0.2155	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-12.35	81.61		0.1514
20		82.80	-12.02	81.61		0.1473
30		140.76	-7.49	81.71		0.0917
40		140.76	-5.14	93.08		0.0553
50	0.19	140.76	-4.50	93.08	0.0013	0.0483
60	0.50	140.76	-4.19	93.08	0.0035	0.0450
70		140.76	-0.92	81.71		0.0113

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-12.07	68.01		0.1774
20		69.00	-11.74	68.01		0.1727
30	30.44	117.30		68.09	0.2595	
40	24.31	117.30		77.57	0.2072	
50	24.59	117.30		77.57	0.2096	
60	24.88	117.30		77.57	0.2121	
70	30.37	117.30		68.09	0.2589	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-12.07	69.35		0.1740
20		69.00	-11.74	69.35		0.1694
30		117.30	-16.55	69.43		0.2384
40		117.30	-12.24	81.69		0.1498
50		117.30	-11.67	81.69		0.1428
60		117.30	-11.37	81.69		0.1392
70		117.30	-10.23	69.43		0.1473

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-12.35	81.61		0.1514
20		82.80	-12.02	81.61		0.1473



30	30.16	140.76	81.71	0.2143
40	24.12	140.76	93.08	0.1714
50	24.48	140.76	93.08	0.1739
60	24.79	140.76	93.08	0.1761
70	30.34	140.76	81.71	0.2155

Absolute Maximum of the all of the Stress Ratio's 0.3808

Governing Element: skirt-1

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
skirt-1	...	-8.9	3.2	19.0	13.3	+ Tens
skirt-1	...	-8.9	-3.2	-19.0	-31.1	- Comp
Skirt-2	...	-8.6	3.2	14.3	9.0	+ Tens
Skirt-2	...	-8.6	-3.2	-14.3	-26.1	- Comp
Bottom Dish	...	-4.0	3.2	11.6	10.8	+ Tens
Bottom Dish	...	-4.0	-3.2	-11.6	-18.9	- Comp
shell 1	...	-2.5	2.5	9.0	9.0	+ Tens
shell 1	...	-2.5	-2.5	-9.0	-13.9	- Comp
shell 2	...	-2.0	2.3	3.8	4.1	+ Tens
shell 2	...	-2.0	-2.3	-3.8	-8.2	- Comp
shell 3	...	-1.8	2.3	2.5	3.1	+ Tens
shell 3	...	-1.8	-2.3	-2.5	-6.6	- Comp
Top dish	...	-0.9	...	0.2	...	+ Tens
Top dish	...	-0.9	...	-0.2	-1.0	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-2

Total vessel (empty) weight	Fg	24342.592 Kg
Head Weight below section 2-2	DeltaFg	4913.815 Kg
Weight of vessel contents (liquid)	Ff	25109.604 Kg
Vessel Gross Weight	Fg + Ff	49452.195 Kg
Skirt Mean Diameter	Dz	2516.000 mm.
Skirt Analysis Thickness (Corroded)	ez	7.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kg
Moment at section 1-1	M1	41126.449 Kg-m.
Applied Moment - Bottom of Skirt	Mb	50783.387 Kg-m.
Weight at Bottom of Skirt	W	49819.066 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	2513.000 mm.
Head Corroded Thickness	eb	7.000 mm.

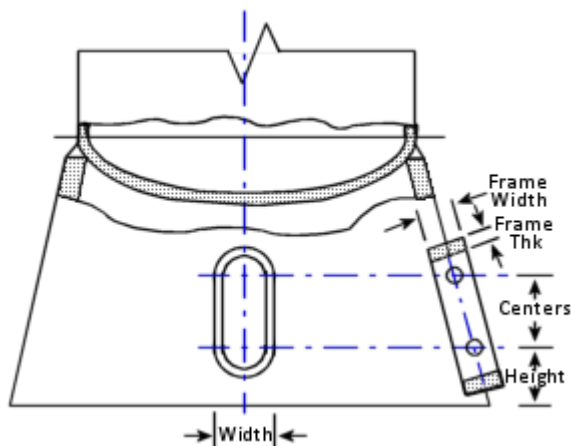
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 628.0 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 2506.0 / (2 * 628.0) = 1.9952$$

$$R = D_i (0.44 * K + 0.02) = 2506.0 * (0.44 * 1.995 + 0.02) = 2250.1326 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 2506.0 * ((0.5 / 1.995) - 0.08) = 427.52 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	4378138.000 cm ⁴
Moment of Inertia	Iyy =	4378138.000 cm ⁴
Section Modulus	W =	34802396.000 mm. ³
Section Metal Area	A =	553.297 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(4378138, 4378138) / 553.297)}$$

$$= 889.54 \text{ mm.}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 889.54/1000.0$$

$$= 0.890$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 41126.445 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -41126.445/.13921E+09 - 49819.066/553.2973$$

$$= -11.727 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +41126.445/.13921E+09 - 49819.066/553.2973$$

$$= -5.933 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 24342.592 - 25109.604 + 4 \cdot 41126.445/2516.0$$

$$= 15930.4 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 24342.592 - 25109.604 - 4 \cdot 41126.445/2516.0$$

$$= -114834.8 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (15930.368 + 4913.815 + 25109.604) / (\pi \cdot 2513.0 \cdot 7.0) + 3.5 \cdot 2513.0 / (4 \cdot 7.0)$$

$$= 39.57 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.1E+06 + 4913.815 + 25109.604) / (\pi \cdot 2513.0 \cdot 7.0) + 3.5 \cdot 2513.0 / (4 \cdot 7.0)$$

$$= 16.36 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (25109.604 + 4913.815) / (\pi \cdot 2513.0 \cdot 7.0) + 3.5 \cdot 2513.0 / (4 \cdot 7.0)$$

$$= 36.74 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 15930.368 / (\pi \cdot 2516.0 \cdot 7.0)$$

$$= 2.82 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -114834.758 / (\pi \cdot 2516.0 \cdot 7.0)$$

$$= -20.35 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(2513.0 + 7.0 - 2516.0 + 7.0) / (427.52 + 7.0)$$

$$= 0.9873$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.987) = 9.126 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(7.0^2 + 7.0^2 + 2 \cdot 7.0 \cdot 7.0 \cdot 0.987)}$$

$$= 6.978 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 6.978 \cdot 15930.4 = 111.16 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 6.978 \cdot -114834.8 = -801.31 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(7.0/7.0)^2$$

$$= 0.573$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot 111.161 / (\pi \cdot 2513.0 \cdot 7.0^2)$$

$$= 9.69 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot -801.311 / (\pi \cdot 2513.0 \cdot 7.0^2)$$

$$= -69.84 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot 111.161 / (\pi \cdot 2516.0 \cdot 7.0^2)$$

$$= 9.68 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot -801.311 / (\pi \cdot 2516.0 \cdot 7.0^2)$$

$$= -69.75 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.276/98.937 = 0.0028$$

$$y = 125 \cdot eb/D_b = 125 \cdot 7.0/2513.0 = 0.3482$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.348 + 3.994 = 4.3418$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (3.5 + 0.0) \cdot 2513.0 ((9.13/40.0) \cdot 4.342 - 1) / (4 \cdot 7.0)$$

$$= -0.30 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 39.569 - 9.688 + -0.297$$

$$= 29.58 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 39.569 + 9.688 - -0.297$$

$$= 49.55 \text{ N./mm}^2$$



Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 16.364 - 69.838 + -0.297$$

$$= 85.90 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 16.364 + -69.838 - -0.297$$

$$= -53.18 \text{ N./mm}^2$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 36.742 + 9.688 + -0.297$$

$$= 46.13 \text{ N./mm}^2$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 36.742 - 9.688 - -0.297$$

$$= 27.35 \text{ N./mm}^2$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 36.742 + -69.838 + -0.297$$

$$= -33.39 \text{ N./mm}^2$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 36.742 - -69.838 - -0.297$$

$$= 106.88 \text{ N./mm}^2$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 2.824 - 9.677$$

$$= -6.85 \text{ N./mm}^2$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 2.824 + 9.677$$

$$= 12.50 \text{ N./mm}^2$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -20.354 - -69.754$$

$$= 49.40 \text{ N./mm}^2$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -20.354 + -69.754$$

$$= -90.11 \text{ N./mm}^2$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (39.569/137.997)^2$$

$$= 406.43 \text{ N./mm}^2$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (16.364/137.997)^2$$

$$= 412.70 \text{ N./mm}^2$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 137.997(3 - (1/1.5)) \cdot (36.742/137.997)^2$$

$$= 407.47 \text{ N./mm}^2$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (36.742/137.997)^2)$$

$$= 407.47 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (2.824/137.997)^2)$$

$$= 413.95 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-20.354/137.997)^2)$$

$$= 411.99 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	29.58	406.43	Pass	46
Stot1po	49.55	406.43	Pass	47
Stot1qi	85.90	412.70	Pass	48
Stot1qo	-53.18	412.70	Pass	49
Stot2pi	46.13	407.47	Pass	50
Stot2po	27.35	407.47	Pass	51
Stot2qi	-33.39	407.47	Pass	52
Stot2qo	106.88	407.47	Pass	53
Stot3pi	-6.85	413.95	Pass	54
Stot3po	12.50	413.95	Pass	55
Stot3qi	49.40	411.99	Pass	56
Stot3qo	-90.11	411.99	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Field before being lifted.
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6653.461 mm.
Center of Gravity of the Liquid	5118.390 mm.
Center of Gravity of the Insulation	6200.000 mm.
Center of Gravity of the Stiffening Rings	6800.000 mm.
Center of Gravity of the Nozzles	6088.531 mm.
Center of Gravity of the Added Weights (Operating)	7077.067 mm.
Center of Gravity of the Added Weights (Empty)	7077.067 mm.
Center of Gravity of Bare Shell New and Cold	4907.359 mm.
Center of Gravity of Bare Shell Corroded	4689.763 mm.
Vessel CG in the Operating Condition	5590.836 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5642.445 mm.
Vessel CG in the Test Condition	6043.941 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	9700.00 mm.
Total vessel weight (No Liquid) Twt	26590.03 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	39885.05 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	9350.00 mm.
Design Reaction force at the tailing lug	16073.48 Kgf
Design Reaction force at the lifting lug	23811.57 Kgf
CG distance during field lifting operation	5642.44 mm.
CG distance from tailing lug	5492.44 mm.
CG distance from the nearer lifting lug/trunnion	3707.55 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	6.43	2500.00	77.84 (UG-23)
Shear	2.06	9070.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	4456.5	0.7
1900.00	20489.8	5.1	6840.5	1.1
4550.00	35587.3	5.9	656.5	0.1
6550.00	33734.9	5.6	6816.5	0.7
9650.00	780.9	0.2	374.1	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0826 at elevation 2500.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : Dump Condensate Reciever (V8302) -7 percent

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Center of Gravity Calculation: Step: 24 1:37pm Jun 20,2026



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--- Plot data successfully generated ...----

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**Skirt Data :**

Skirt Outside Diameter at Base	SOD, D6	2523.0000	mm.
Skirt Thickness	STHK	8.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	114.00	N./mm ²
Basing Yield Stress		248.00	N./mm ²
Inside Diameter of Basing	DI, D4	2343.0000	mm.
Outside Diameter of Basing	DOU, D5	2803.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Root Area of a Single Bolt	Area	8.6100	cm ²
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	48.30	N./mm ²
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	2663.0000	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.4	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	130.0000	mm.
Width of Gussets at Base Plate	BWDT	130.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	248.0	N./mm ²
Height of Chair	h1	300.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	26590.0	Kgf
Operating Weight of Vessel	ROW	51699.6	Kgf
Earthquake Moment on Basing	EQMOM	75094.3	Kg-m.
Wind Moment on Basing	WIMOM	24450.4	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 75094/2663.0) - 50078) / 16$$



= 3919.8 Kgf

Bolt Load, Windload Empty Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 24450/2663.0) - 26590) / 16$

= 633.5 Kgf

Bolt Load, Windload Vortex Shedding Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 0/2663.0) - 51700) / 16$

= -3231.2 Kgf

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 8.6100 cm²

Area Available in all the Bolts Abss * RN : 137.7600 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.406	0.203	1.779	2.211	0.414	0.784	9497.6	59575.9	1263.9
4	0.304	0.355	1.521	2.433	0.437	0.781	8099.7	58178.0	979.7
6	0.380	0.368	1.717	2.267	0.420	0.784	9141.8	59220.2	1186.7
8	0.374	0.371	1.701	2.281	0.421	0.784	9053.6	59131.9	1168.1
10	0.369	0.370	1.689	2.291	0.422	0.784	8987.6	59065.9	1154.3
12	0.370	0.370	1.692	2.289	0.422	0.784	9004.1	59082.4	1157.7
13	0.370	0.370	1.691	2.289	0.422	0.784	9001.3	59079.7	1157.2

The Actual Stress in a Single Bolt [Sbolt]:

= $2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$

= $2 \cdot 9001.3 / (1.647 \cdot 2663.0 \cdot 2.289)$

= 17.589 N./mm², Should be ≤ 48.3 N./mm²

Thickness of the Band of Bolting Steel [T1]

= $RN \cdot Bolt\ Area / (\pi \cdot Dc)$

= $16 \cdot 8.61 / (\pi \cdot 2663.0)$

= 1.647 mm.

Check the Bearing Stress in the Concrete [fc(max)]

= $fc[(2kd + t3) / (2kd)]$

= $1.052[(2 \cdot 0.3703 \cdot 2663.0 + 230.0) / (2 \cdot 0.3703 \cdot 2663.0)]$

= 1.175 N./mm², Should be ≤ 22.5 N./mm²

Values for table 10.3, l = 140.0, b = 261.439, l/b = 0.535497

Maximum Moment per unit width [Mmax]:

= $\max(Mx, My) = \max(286.098, 702.93) = 702.93$ Kgf

Reqd Thickness of Basing, Brownell & Young Method [T]:

= $\sqrt{(6 \cdot Mmax / fallow)} + Ca$

= $\sqrt{(6 \cdot 702.93 / 163.7)} + 1.0$

= 16.897 mm.

Nomenclature:

a	= (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P	= Sa * Abss	Maximum Load on one Bolt
l		Radial Top Plate Width
lavg		Average Gusset Plate Width
g1	= Gamma 1	Constant Term f(b/l)
g2	= Gamma 2	Constant Term f(b/l)
g	= Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb		Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 90.0, b/l = 0.642857

As b/l (0.643) is less than 1, inverting b/l = 1.556.

Moment Term, based on geometry [Mo]:

= $P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1)] - [(0.7 - g2)P / (4\pi)]$

= $4240.47 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0 / 140.0)) /$

$(\pi \cdot 34.925))) + 1)] - [(0.7 - 0.065) \cdot 4240.47 / (4 \cdot \pi)]$

= 533.9864 Kgf



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(533.99)}{165.33}} + 1.0$$

$$= 14.7857 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{\text{avg}}) + C_a$$

$$= 4240.47 / (124.11 \cdot 130.0) + 1.0$$

$$= 10.525 \text{ (not less than } 9.525 + 1.0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 2663.0 / 16 = 522.879 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76 (4240 \cdot 70.0 / (522.88 \cdot 262.0 \cdot 171))^{2/3} \cdot 1261.51^{1/3} + C_a$$

$$= 4.736 + 1.0 = 5.736 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9727.8 / (8.61 \cdot 16)$$

$$= 6.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	16.8965	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	14.7857	mm.
Actual Top Ring Thickness as entered by user	38.0000	mm.
Required Gusset thickness, + CA	10.5250	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	5.7356	mm.
Given Thickness of Skirt	8.0000	mm.
Required Gusset Height to meet local stress	113.7863	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 31.077 \cdot (8.0 - 1.0)$$

$$= 22.18 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 22 / [(0.4 \cdot 215) \cdot 2 \cdot 0.707]$$

$$= 1.786 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cm}w_{th} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 4240.5 / (181.0 + 2 \cdot (230.0 + 38.0))$$

$$= 5.914 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cm}w_{th} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 4240.5 \cdot 70.0 / (181.0 \cdot (268.0) + 0.6667 \cdot (268.0)^2)$$

$$= 3.079 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{5.91^2 + 3.08^2}$$

$$= 6.668 \text{ Kg/mm.}$$



Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 6.67 / [(0.4 \cdot 215) \cdot 2 \cdot 0.707]$$

$$= 0.537 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 4240.5 / (2 \cdot 130.0)$$

$$= 16.309 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 4240.5 \cdot 70.0 / (2 \cdot 268.0 \cdot 130.0)$$

$$= 4.26 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{(W_v^2 + W_h^2)}$$

$$= \sqrt{(16.31^2 + 4.26^2)}$$

$$= 16.857 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 16.86 / [(0.4 \cdot 215) \cdot 2 \cdot 0.707]$$

$$= 1.357 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	1.3569	mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Single		
Tail lug yield stress	Syt	248.00	N./mm ²
Tail lug allowable stress	Sat	114.00	N./mm ²
Tail lug thickness	TTL	34.0000	mm.
Pin hole diameter	DPH	45.0000	mm.
Fillet weld size	WS	20.0000	mm.
Tail lug load	TLOAD	16073	Kgf

Decay Length Along the Skirt (e):

$$= 0.5 \cdot \sqrt{(DS/2 \cdot T_s)}$$

$$= 0.5 \cdot \sqrt{(2523.0/2 \cdot 8.0)}$$

$$= 50.23 \text{ mm.}$$

Skirt Effective Length :

$$= HG + TTA + e$$

$$= 230.0 + 38.0 + 50.23$$

$$= 318.23 \text{ mm.}$$

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm ²	Moment Area mm. ³
Top plate, t	160.00	140.00	53.20	851200.06
Skirt, s	86.00	318.23	25.46	218941.95
Bottom plate, b	115.00	230.00	73.60	846400.00
TOTAL			152.26	1916541.88

Calculated Overall CG (Y):

$$= M_{\text{total}} / A_{\text{total}}$$



$$= 1916542/152.26$$

$$= 125.87 \text{ mm.}$$

Calculated Moment of Inertia (I):

$$= (At \cdot Yt^2 + Topwth^3 \cdot TTA/12 + As \cdot Ys^2 + Ts^3 \cdot Hs/12 + Ab \cdot Yb^2 + Hb^3 \cdot Tb/12) - Mtotal^2/Atotal$$

$$= (53.2 \cdot 160.0^2 + 140.0^3 \cdot 38.0/12 + 25.46 \cdot 86.0^2 + 8.0^3 \cdot 318.23/12 + 73.6 \cdot 115.0^2 + 230.0^3 \cdot 32.0/12) - 1916542^2/152.26$$

$$= 5226.18 \text{ cm}^4$$

Required Section Modulus (Zr) - per Roark's Table 17 ref 20

$$= TLOAD \cdot (DI/2 + Y) \cdot (K2 + 0.5) / (0.66 \cdot Sy \cdot 2 \cdot \pi)$$

$$= 16073.5 \cdot (2343.0/2 + 125.87) \cdot (1.0 + 0.5) / (0.66 \cdot 248.0 \cdot 2 \cdot 3.142)$$

$$= 298282.28 \text{ mm.}^3$$

Calculated Section Modulus (Zc):

$$= I / Y$$

$$= 5226.18/125.87$$

$$= 415190.72 \text{ mm.}^3$$

Allowable Shear at the Pin Hole (Ssa):

$$= 0.4 \cdot Syt$$

$$= 0.4 \cdot 248.0$$

$$= 99.20 \text{ N./mm}^2$$

Tail Lug Radius (R):

$$= (TBA + HG + TTA) / 2$$

$$= (32.0 + 230.0 + 38.0) / 2$$

$$= 150.00 \text{ mm.}$$

Calculated Shear at the Pin Hole (Ssc):

$$= TLOAD / (2 \cdot TTL \cdot (R - DPH / 2))$$

$$= 16073.5 / (2 \cdot 34.0 \cdot (150.0 - 45.0/2))$$

$$= 18.18 \text{ N./mm}^2$$

Allowable Bearing Stress at the Pin Hole (Sba):

$$= 0.75 \cdot Syt$$

$$= 0.75 \cdot 248.0$$

$$= 186.00 \text{ N./mm}^2$$

Calculated Bearing Stress at the Pin Hole (Sbc):

$$= TLOAD / (DPH \cdot TTL)$$

$$= 16073.5 / (45.0 \cdot 34.0)$$

$$= 103.03 \text{ N./mm}^2$$

Weld Length (LW):

$$= 2 \cdot (TOPWTH + TBA + TTA + HG + (DOU - DS) / 2)$$

$$= 2 \cdot (140.0 + 32.0 + 38.0 + 230.0 + (2803.0 - 2523.0)/2)$$

$$= 1143.00 \text{ mm.}$$

Allowable Weld Stress (Swa):

$$= 0.49 \cdot Sat$$

$$= 0.49 \cdot 114.0$$

$$= 55.86 \text{ N./mm}^2$$

Calculated Weld Stress (Swc):

$$= TLOAD / (0.707 \cdot WS \cdot LW)$$

$$= 16073.5 / (0.707 \cdot 20.0 \cdot 1143.0)$$

$$= 9.75 \text{ N./mm}^2$$

Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
Section Modulus	415190.72	298282.28	Passed
Shear Stress	18.18	99.20	Passed
Bearing Stress	103.03	186.00	Passed

LP Condensate Flash Vessel

PV Elite 28 U1 Licensee: SL User

FileName : Dump Condensate Reciever (V8302) -7 percent

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Basering Calculations:

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Weld Stress	9.75	55.86	Passed
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Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: N2-6"****From: 30**

Pressure for Reinforcement Calculations	P	6.568	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	300.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgmv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		75.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		270.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1200.0	mm.
Centerline Distance to Tangent		1000.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N2-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 6.625 in.
 Actual Thickness Used in Calculation 0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.57 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.57)$
 $= 5.3580$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.57 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 6.57)$
 $= 0.4673$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 281.2740 mm.
 Parallel to Vessel Wall, opening length d 140.6370 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 17.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.758	4.724	NA
Area in Shell	A1	2.241	0.649	NA
Area in Nozzle Wall	A2	4.175	4.141	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.802	0.802	NA
Area in Element	A5	11.170	11.170	NA
TOTAL AREA AVAILABLE	Atot	18.388	16.761	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 82.90 degs.



The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	173.6829	10.0000 mm.
Based on given Pad Diameter:	318.2750	0.4841 mm.
Based on Shell or Nozzle Thickness:	174.9816	10.0000 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (140.637 \cdot 5.358 \cdot 1.0 + 2 \cdot 14.3575 \cdot 5.358 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 7.758 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 140.637(1.0 \cdot 7.0 - 1.0 \cdot 5.358) - 2 \cdot 14.358$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 5.358) \cdot (1 - 0.855)$$

$$= 2.241 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$

$$= (2 \cdot 17.5) \cdot (14.36 - 0.47) \cdot 0.8551 / \sin(84.7)$$

$$= 4.175 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (0.9375) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.802 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (281.274 - 169.5737) \cdot 10.0 \cdot 1.0$$

$$= 11.170 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.958, tr = 0.467, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.765, Temp. Reduction = 13 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.765, Temp. Reduction = 13 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, tr = 5.358, c = 3.0 mm., E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.765$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 5.358, c = 3.0 mm., $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.765$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Gov. MDMT of the Nozzle : -42 °C
 Gov. MDMT of the Reinforcement Pad : -42 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.57/51.00 = 0.129$

Weld Size Calculations, Description: N2-6"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$4.5000 = 0.5 * \text{TminPad}$	$5.6560 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (7.7583 - 2.241 + 2 \cdot 14.3575 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 5.358)) \cdot 138)$
 $= 8331.08 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.1749 + 11.17 + 0.8016 - 0.0 \cdot 0.86) \cdot 138$
 $= 22720.76 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.1749 + 0.0 + 0.8551 + (1.7187)) \cdot 138$
 $= 9496.52 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.1749 + 0.0 + 0.8016 + 11.17 + (1.7187)) \cdot 138$
 $= 25139.30 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 169.5737 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15704. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$



$$= (3.1416/2.0) \cdot 318.275 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 27577. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 77.5527) \cdot (15.9576 - 1.6) \cdot 0.7 \cdot 118$$

$$= 29463. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 169.5737 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 27737. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 169.5737 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 19416. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (27577 + 29463) = 57040 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15704 + 27737 + 19416 + 0) = 62857 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (27577 + 19416 + 0) = 46993 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 57039	Kgf, must exceed W = 8331 Kg	or W1 = 22720 Kg
Path 2-2 = 62856	Kgf, must exceed W = 8331 Kg	or W2 = 9496 Kg
Path 3-3 = 46992	Kgf, must exceed W = 8331 Kg	or W3 = 25139 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

Note : Checking Nozzle in the Latitudinal direction**Reinforcement CALCULATION, Description: N2-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 6.57 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 6.57)$$

$$= 5.3580 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 6.57 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 6.57)$$

$$= 0.4673 \text{ mm.}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]



Parallel to Vessel Wall (Diameter Limit) D1 279.1198 mm.
 Parallel to Vessel Wall, opening length d 139.5599 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 17.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.701	4.689	NA
Area in Shell	A1	2.223	0.643	NA
Area in Nozzle Wall	A2	4.157	4.123	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.802	0.802	NA
Area in Element	A5	11.084	11.084	NA
TOTAL AREA AVAILABLE	Atot	18.266	16.653	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness:	173.4617	10.0000 mm.
Based on given Pad Diameter:	318.2750	0.4679 mm.
Based on Shell or Nozzle Thickness:	173.4617	10.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (139.5599 \cdot 5.358 \cdot 1.0 + 2 \cdot 14.3575 \cdot 5.358 \cdot 1.0 \cdot (1 - 0.86))$
 $= 7.701 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 139.56 (1.0 \cdot 7.0 - 1.0 \cdot 5.358) - 2 \cdot 14.358$
 $(1.0 \cdot 7.0 - 1.0 \cdot 5.358) \cdot (1 - 0.855)$
 $= 2.223 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 17.5) \cdot (14.36 - 0.47) \cdot 0.8551$
 $= 4.157 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$
 $= (0.9375) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 0.802 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
 $= (279.1198 - 168.275) \cdot 10.0 \cdot 1.0$
 $= 11.084 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.1805 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.9479 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.9479 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.9362 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8200 mm.



Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.82, \max(8.9479, 5.5)]$$

$$= 7.8200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.1805, 7.82)$$

$$= 7.8200 \text{ mm.}$$

Available Nozzle Neck Thickness = 15.9576 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	277.4 N./mm ²	Passed
Occasional	:	1.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	13.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N2-6"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm.
Intermediate Calc. for pad/shell Welds	TminPad	9.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	4.5000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (7.7006 - 2.2233 + 2 \cdot 14.3575 \cdot 0.8551 \cdot (1.0 \cdot 7.0 - 5.358)) \cdot 138)$$

$$= 8274.76 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (4.157 + 11.0845 + 0.8016 - 0.0 \cdot 0.86) \cdot 138$$

$$= 22575.18 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (4.157 + 0.0 + 0.8551 + (1.7187)) \cdot 138$$

$$= 9471.32 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (4.157 + 0.0 + 0.8016 + 11.0845 + (1.7187)) \cdot 138$$

$$= 24993.72 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 318.275 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 27577. \text{ Kg}$$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 76.9587) \cdot (15.9576 - 1.6) \cdot 0.7 \cdot 118$$

$$= 29237. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 168.275 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 19267. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (27577 + 29237) = 56814 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 27524 + 19267 + 0) = 62375 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (27577 + 19267 + 0) = 46844 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 56814	Kgf, must exceed W = 8274 Kg	or W1 = 22575 Kg
Path 2-2 = 62375	Kgf, must exceed W = 8274 Kg	or W2 = 9471 Kg
Path 3-3 = 46844	Kgf, must exceed W = 8274 Kg	or W3 = 24993 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 12.8973 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9864 mm.

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N2-6" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	4500.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	180.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type Type Round



WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.6000 mm.
Nozzle Diameter	Dn	168.275 mm.
Nozzle Thickness	Tn	15.958 mm.
Nozzle Material		SA-106 B
Nozzle UNS Number		K03006
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²
Thickness of Reinforcing Pad	Tpad	10.000 mm.
Diameter of Reinforcing Pad	Dpad	318.275 mm.
Design Internal Pressure	Dp	6.568 bars
Include Pressure Thrust		Yes

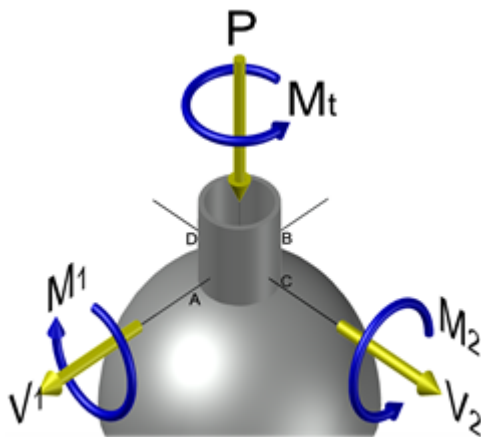
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-1125.0	Kgf
Longitudinal Shear (SUS)	(V1) V1	750.0	Kgf
Circumferential Shear (SUS)	(Vc) V2	750.0	Kgf
Circumferential Moment (SUS)	(Mc) M1	315.0	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	315.0	Kg-m.
Torsional Moment (SUS)	Mt	450.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2256.5 (10.0 - 3.0))} \\
 &= 583.019 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-2103.1	Kgf
Circumferential Shear	(VC) V2	750.0	Kgf
Longitudinal Shear	(VL) V1	750.0	Kgf
Circumferential Moment	(MC) M1	315.0	Kg-m.
Longitudinal Moment	(ML) M2	315.0	Kg-m.



Torsional Moment MT 450.0 Kg-m.

Dimensionless Parameters: U = 0.43 TAU = 5.36 RHO = 1.18

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.08100	(A,B,C,D)
M(x) / P	SP 2	0.09314	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.16855	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.25430	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.16855	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.25430	(A,B,C,D)
N(y) * T / P	SP 2	0.16150	(A,B,C,D)
M(y) / P	SP 2	0.04206	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.08875	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.13946	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.08875	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.13946	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8
Rad.	Bend. P	39.9	-39.9	39.9	-39.9	39.9	-39.9	39.9	-39.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-9.2	-9.2	9.2	9.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-83.2	83.2	83.2	-83.2
Rad.	Memb. ML	-9.2	-9.2	9.2	9.2	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-83.2	83.2	83.2	-83.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-46.7	39.9	138.0	-108.1	-46.7	39.9	138.0	-108.1
Tang.	Memb. P	11.5	11.5	11.5	11.5	11.5	11.5	11.5	11.5
Tang.	Bend. P	18.0	-18.0	18.0	-18.0	18.0	-18.0	18.0	-18.0
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-4.8	-4.8	4.8	4.8
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-45.6	45.6	45.6	-45.6
Tang.	Memb. ML	-4.8	-4.8	4.8	4.8	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-45.6	45.6	45.6	-45.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-20.9	34.3	80.0	-47.3	-20.9	34.3	80.0	-47.3
Shear VC		1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Shear MT		5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8
Tot. Shear		7.5	7.5	4.2	4.2	4.2	4.2	7.5	7.5
Str. Int.		48.7	45.1	138.3	108.4	47.4	42.1	139.0	109.0

Unitless Prm: U = 1.27 TAU = 0.00 (10.58) RHO = 0.00 (0.49)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.04802	(A,B,C,D)
M(x) / P	SR 2	0.02452	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.05265	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.04268	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.05265	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.04268	(A,B,C,D)



N(y) * T / P	SR 2	0.01447	(A,B,C,D)
M(y) / P	SR 2	0.00737	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.01579	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.01289	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.01579	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.01289	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		20.2	20.2	20.2	20.2	20.2	20.2	20.2	20.2
Rad. Bend. P		61.9	-61.9	61.9	-61.9	61.9	-61.9	61.9	-61.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-26.4	-26.4	26.4	26.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-128.5	128.5	128.5	-128.5
Rad. Memb. ML		-26.4	-26.4	26.4	26.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-128.5	128.5	128.5	-128.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-72.7	60.3	237.0	-143.7	-72.7	60.3	237.0	-143.7
Tang. Memb. P		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tang. Bend. P		18.6	-18.6	18.6	-18.6	18.6	-18.6	18.6	-18.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.9	-7.9	7.9	7.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-38.8	38.8	38.8	-38.8
Tang. Memb. ML		-7.9	-7.9	7.9	7.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-38.8	38.8	38.8	-38.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-22.0	18.4	71.4	-43.4	-22.0	18.4	71.4	-43.4
Shear VC		2.1	2.1	-2.1	-2.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear MT		4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Tot. Shear		6.1	6.1	1.9	1.9	1.9	1.9	6.1	6.1
Str. Int.		73.4	61.2	237.0	143.8	72.8	60.4	237.2	144.1

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Rad. Pl (SUS)		-3.4	-3.4	15.0	15.0	-3.4	-3.4	15.0	15.0
Rad. Q (SUS)		-43.3	43.3	123.1	-123.1	-43.3	43.3	123.1	-123.1
Long. Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Long. Pl (SUS)		6.7	6.7	16.4	16.4	6.7	6.7	16.4	16.4
Long. Q (SUS)		-27.6	27.6	63.6	-63.6	-27.6	27.6	63.6	-63.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.6	1.6	-1.6	-1.6	-1.6	-1.6	1.6	1.6
Shear Q (SUS)		5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8
Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Pm+Pl (SUS)		50.5	50.5	61.0	61.0	50.5	50.5	61.0	61.0
Pm+Pl+Q (Total)		29.8	88.6	181.9	64.9	27.1	85.7	182.5	65.5

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.53	138.00	Passed
Pm+Pl (SUS)	60.97	207.00	Passed
Pm+Pl+Q (TOTAL)	182.50	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Rad.	Pl (SUS)	-6.2	-6.2	46.6	46.6	-6.2	-6.2	46.6	46.6
Rad.	Q (SUS)	-66.5	66.5	190.4	-190.4	-66.5	66.5	190.4	-190.4
Long.	Pm (SUS)	105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Long.	Pl (SUS)	-1.8	-1.8	14.0	14.0	-1.8	-1.8	14.0	14.0
Long.	Q (SUS)	-20.2	20.2	57.4	-57.4	-20.2	20.2	57.4	-57.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.1	2.1	-2.1	-2.1	-2.1	-2.1	2.1	2.1
Shear	Q (SUS)	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Pm (SUS)		105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Pm+Pl (SUS)		104.7	104.7	152.5	152.5	104.7	104.7	152.5	152.5
Pm+Pl+Q (Total)		84.4	166.9	342.7	100.4	83.8	166.1	342.9	101.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	105.71	138.00	Passed
Pm+Pl (SUS)	152.46	207.00	Passed
Pm+Pl+Q (TOTAL)	342.92	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: D1-3"****From: 30**

Pressure for Reinforcement Calculations	P	6.569	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	200.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	190.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		50.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		1200.0	mm.
Centerline Distance to Tangent		900.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: D1-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.57 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.57)$
 $= 5.3588$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.57 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 6.57)$
 $= 0.2469$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.7422 mm.
Parallel to Vessel Wall, opening length	d	72.8711 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		17.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.735, tr = 0.247, c = 1.6 mm., $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.735, tr = 0.247, c = 1.6 mm., $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.735, tr = 0.247, c = 1.6 mm., $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, tr = 5.359, c = 3.0 mm., $E* = 1.0$
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.766$, Temp. Reduction = 13 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.735$, $t_r = 0.247$, $c = 1.6$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -42 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.57/51.00 = 0.129$

Weld Size Calculations, Description: D1-3"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.1346 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$4.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: D1-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, t_r [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.57 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.57)$
 $= 5.3588$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, t_{rn} [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.57 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 6.57)$
 $= 0.2469$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2618 mm.
Parallel to Vessel Wall, opening length	d	72.6309 mm.
Normal to Vessel Wall (Thickness Limit), pad side T_{lwp}		17.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1-3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.9989$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 5.5000$ mm.
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.9488$ mm.
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.9488$ mm.
 Wall Thickness, shell/head, external pressure $trb_2 = 3.9362$ mm.
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 5.5000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 6.4000$ mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.4, \max(8.9488, 5.5)]$
 $= 6.4000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(1.9989, 6.4)$
 $= 6.4000$ mm.

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	257.6 N./mm ²	Passed
Occasional	:	1.3,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	25.1,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: D1-3"

Intermediate Calc. for nozzle/shell Welds	T_{min}	8.1346	mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	9.0000	mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$4.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.3962 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.4355 mm.

*Warning:
 The equations for nozzle local stress analysis pertain to radial nozzles only
 and this nozzle is not specified as radial. Please check the specific
 analysis code for all assumptions and limitations.*

Input Echo, WRC107/537 Item 1, Description: D1-3" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	4500.000	mm.
Vessel Thickness	Tv	10.000	mm.
Design Temperature	T1	180.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

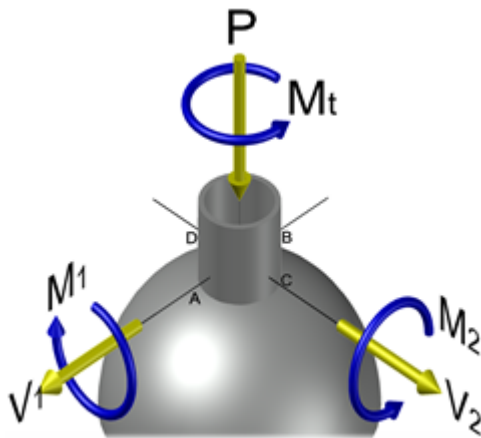
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	88.900	mm.
Nozzle Thickness	Tn	9.735	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Diameter of Reinforcing Pad	Dpad	190.000	mm.
Design Internal Pressure	Dp	6.569	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	611.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	407.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	407.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	91.3	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	91.3	Kg-m.
Torsional Moment (SUS)	Mt	130.5	Kg-m.

Use Interactive Control	No	
WRC107 Version	Version	March 1979
Include Pressure Stress Indices per Div. 2	No	
Compute Pressure Stress per WRC-368	No	
Local Loads applied at end of Nozzle/Attachment	No	

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(2256.5 (10.0 - 3.0))}$$

$$= 503.645 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	611.9	Kgf
Circumferential Shear	(VC)	V2	407.9	Kgf
Longitudinal Shear	(VL)	V1	407.9	Kgf
Circumferential Moment	(MC)	M1	91.3	Kg-m.
Longitudinal Moment	(ML)	M2	91.3	Kg-m.
Torsional Moment		MT	130.5	Kg-m.

Dimensionless Param: U = 0.23 TAU = 5.00 (4.96) RHO = 2.09

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.12629	(A,B,C,D)
$M(x) / P$	SP 3	0.09547	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.36941	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.31675	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.36941	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.31675	(A,B,C,D)
$N(y) * T / P$	SP 3	0.23685	(A,B,C,D)
$M(y) / P$	SP 3	0.10303	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.08904	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	0.51369	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.08904	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	0.51369	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6
Rad.	Bend. P	-11.9	11.9	-11.9	11.9	-11.9	11.9	-11.9	11.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-5.8	-5.8	5.8	5.8
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-30.0	30.0	30.0	-30.0
Rad.	Memb. ML	-5.8	-5.8	5.8	5.8	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-30.0	30.0	30.0	-30.0	0.0	0.0	0.0	0.0

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-11.2	-11.2	-11.2	-11.2	-11.2	-11.2	-11.2	-11.2
Rad.	Bend. P	-37.0	37.0	-37.0	37.0	-37.0	37.0	-37.0	37.0
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-15.5	-15.5	15.5	15.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-109.3	109.3	109.3	-109.3
Rad.	Memb. ML	-15.5	-15.5	15.5	15.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-109.3	109.3	109.3	-109.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-173.0	119.6	76.6	-68.1	-173.0	119.6	76.6	-68.1
Tang. Memb. P		-3.4	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4	-3.4
Tang.	Bend. P	-11.4	11.4	-11.4	11.4	-11.4	11.4	-11.4	11.4
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-32.8	32.8	32.8	-32.8
Tang.	Memb. ML	-4.7	-4.7	4.7	4.7	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-32.8	32.8	32.8	-32.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-52.2	36.1	22.6	-20.0	-52.2	36.1	22.6	-20.0
Shear VC		1.9	1.9	-1.9	-1.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Shear MT		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2



Tot. Shear	5.1	5.1	1.3	1.3	1.3	1.3	5.1	5.1
Str. Int.	173.2	119.9	76.7	68.2	173.0	119.6	77.1	68.7

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Rad. Pl (SUS)		-8.5	-8.5	3.2	3.2	-8.5	-8.5	3.2	3.2
Rad. Q (SUS)		-41.9	41.9	18.1	-18.1	-41.9	41.9	18.1	-18.1
Long. Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Long. Pl (SUS)		-6.3	-6.3	-3.5	-3.5	-6.3	-6.3	-3.5	-3.5
Long. Q (SUS)		-61.6	61.6	35.9	-35.9	-61.6	61.6	35.9	-35.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.7	1.7	-1.7	-1.7	-1.7	-1.7	1.7	1.7
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		43.5	43.5	43.5	43.5	43.5	43.5	43.5	43.5
Pm+Pl (SUS)		38.1	38.1	47.1	47.1	38.1	38.1	47.1	47.1
Pm+Pl+Q (Total)		27.3	101.2	77.4	29.4	25.4	99.6	79.9	30.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43.53	138.00	Passed
Pm+Pl (SUS)	47.15	207.00	Passed
Pm+Pl+Q (TOTAL)	101.24	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Rad. Pl (SUS)		-26.7	-26.7	4.2	4.2	-26.7	-26.7	4.2	4.2
Rad. Q (SUS)		-146.3	146.3	72.4	-72.4	-146.3	146.3	72.4	-72.4
Long. Pm (SUS)		105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Long. Pl (SUS)		-8.0	-8.0	1.3	1.3	-8.0	-8.0	1.3	1.3
Long. Q (SUS)		-44.2	44.2	21.3	-21.3	-44.2	44.2	21.3	-21.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.9	1.9	-1.9	-1.9	-1.9	-1.9	1.9	1.9
Shear Q (SUS)		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Pm (SUS)		105.7	105.7	105.7	105.7	105.7	105.7	105.7	105.7
Pm+Pl (SUS)		97.9	97.9	110.9	110.9	97.9	97.9	110.9	110.9
Pm+Pl+Q (Total)		121.2	225.7	182.4	85.7	120.8	225.4	182.8	86.2

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	105.72	138.00	Passed
Pm+Pl (SUS)	110.91	207.00	Passed
Pm+Pl+Q (TOTAL)	225.65	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: M1-24"****From: 40**

Pressure for Reinforcement Calculations	P	6.366	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2858.3330	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	909.6000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		150.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.37 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.37)$
 $= 5.7962$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.37 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 6.37)$
 $= 1.4035$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.4576 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1183.2001 mm.
 Parallel to Vessel Wall, opening length d 591.6000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 22.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	34.290	25.273	NA
Area in Shell	A1	18.954	2.698	NA
Area in Nozzle Wall	A2	3.418	3.394	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.640	1.640	NA
Area in Element	A5	36.000	36.000	NA
TOTAL AREA AVAILABLE	Atot	60.012	43.733	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 755.7687 12.0000 mm.
 Based on given Pad Diameter: 909.6000 5.8467 mm.
 Based on Shell or Nozzle Thickness: 755.7687 12.0000 mm.



Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(591.6 \cdot 8.5439 \cdot 1 + 2 \cdot 9.0 \cdot 8.5439 \cdot 1(1 - 1.0))$$

$$= 25.273 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 591.6(1.0 \cdot 9.0 - 1.0 \cdot 8.544) - 2 \cdot 9.0$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 8.5439) \cdot (1 - 1.0)$$

$$= 2.698 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.5) \cdot (9.0 - 1.46) \cdot 1.0$$

$$= 3.394 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 8.0^2 \cdot 1.0$$

$$= 1.640 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (909.6 - 609.6) \cdot 12.0 \cdot 1.0$$

$$= 36.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 4.4576 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 4.4576 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } tg = 12.0, tr = 1.403, c = 3.0 \text{ mm., } E* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.156, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } tg = 12.0, tr = 1.403, c = 3.0 \text{ mm., } E* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.156, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } tg = 12.0, tr = 1.403, c = 3.0 \text{ mm., } E* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.156, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

$$\text{Govrn. thk, } tg = 12.0, tr = 5.796, c = 3.0 \text{ mm., } E* = 1.0$$



Thickness Ratio = $tr * E / (tg - c) = 0.644$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $tg = 12.0$, $tr = 5.796$, $c = 3.0$ mm., $E = 1.0$
 Thickness Ratio = $tr * E / (tg - c) = 0.644$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -43 °C
 Gov. MDMT of the Reinforcement Pad : -43 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.37/19.60 = 0.325$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000$ mm.
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 9.0000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$
Pad Weld	$4.5000 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (25.2728 - 2.6985 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 9.0 - 8.5439)) \cdot 138)$
 $= 31881.13 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / 707)^2 \cdot fr2) \cdot Sv$
 $= (3.3941 + 36.0 + 1.64 - 0.0 \cdot 1.0) \cdot 138$
 $= 57741.37 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (3.3941 + 0.0 + 1. + (1.62)) \cdot 138$
 $= 8462.73 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (3.3941 + 0.0 + 1.64 + 36.0 + (1.62)) \cdot 138$
 $= 60020.97 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66024. \text{ Kg}$



Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 909.6 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 78813. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 300.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 83635. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 609.6 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 119652. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 89739. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (78813 + 83635) = 162448 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66024 + 119652 + 89739 + 0) = 275415 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (78813 + 89739 + 0) = 168552 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 162448 Kgf, must exceed W = 31881 Kgf	or W1 = 57741 Kgf
Path 2-2 = 275415 Kgf, must exceed W = 31881 Kgf	or W2 = 8462 Kgf
Path 3-3 = 168552 Kgf, must exceed W = 31881 Kgf	or W3 = 60020 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 37.7306 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 349.7307 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.008 \%$$

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**Input, Nozzle Desc: K1-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.361	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2858.3330	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1600.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.36 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.36)$
 $= 5.7919$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.36 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 6.36)$
 $= 0.1313$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3924	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.7919	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.7919	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.9395	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(8.7919, 5.5)]$
 $= 7.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.3924, 7.8)$
 $= 7.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 12.0, tr = 5.792, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.644$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6.36 / 51.00 = 0.125$

Weld Size Calculations, Description: K1-2"



Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7113 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0863 mm.

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**Input, Nozzle Desc: K2-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.483	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2858.3330	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.48 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.48)$
 $= 5.9034 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.48 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 6.48)$
 $= 0.1338 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3924 mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.9034 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.9034 mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.9395 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm.
Wall Thickness per table UG-45	tb3	= 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(8.9034, 5.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.3924, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 12.0, tr = 5.903, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.656$, Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6.48 / 51.00 = 0.127$

Weld Size Calculations, Description: K2-2"



Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7113 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.0863 mm.

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**Input, Nozzle Desc: N5-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.483	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2858.3330	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.382	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.48 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.48)$
 $= 5.9034$ mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.48 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 6.48)$
 $= 0.1654$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3185 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	93.8896	mm.
Parallel to Vessel Wall, opening length	d	46.9448	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	16.7253	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N5-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3185	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.9034	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.9034	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.9395	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(8.9034, 5.5)]$
 $= 6.4200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3185, 6.42)$
 $= 6.4200$ mm.

Available Nozzle Neck Thickness = 9.6901 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	40.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	254.4 N./mm ²	Passed
Occasional	:	1.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	28.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.165, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.165, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 6.48/51.00 = 0.127

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.6901 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.6831 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3639 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.3640 mm.

Input Echo, WRC297 Item 1, Description: N5-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2500.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	60.325	mm.
Nozzle Thickness	9.6901	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	6.48	bars
Radial Load	P	382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



Warning - The ratio, D_n/T_n (6.225) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(1257.5(12.0 - 3.0))}$$

$$= 411.391 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2515.000 mm.
Nozzle Outside Diameter	(d) =	60.325 mm.
Vessel Thickness used	(T) =	9.000 mm.
Nozzle Thickness used	(t) =	6.690 mm.
T / t	=	1.345
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.401$$

Nr/P	=	0.203
Mr/P	=	0.278
M0/P	=	0.117
N0/P	=	0.180
MrD/Mc	=	0.270
NrDL/Mc	=	0.162
M0d/Mc	=	0.139
N0DL/Mc	=	-0.112
MrD/Ml	=	0.254
NrDL/Ml	=	0.160
M0D/Ml	=	0.129
N0DL/Ml	=	-0.039

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-32	32	-32	32
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	2	2	-2	-2
Outplane Bending (ML)	-55	55	55	-55
Normal Pressure Stress	89	90	89	90
Outplane Stress Summary	-3	172	101	56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-9	-9	-9	-9
Inplane Bending (P)	-77	77	-77	77
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-109	109	109	-109
Inplane Pressure Stress	44	44	44	44
Inplane Stress Summary	-162	210	78	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	9	9	9	9

Shear Stress Summary	12	12	6	6
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	163	214	103	57
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-8	-8	-8	-8
Outplane Bending (P)	-32	32	-32	32
Outplane Membrane (Mc)	7	7	-7	-7
Outplane Bending (Mc)	-59	59	59	-59
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44

Outplane Stress Summary	-47	136	56	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-9	-9	-9	-9
Inplane Bending (P)	-77	77	-77	77
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-116	116	116	-116
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	89	90	89	90

Inplane Stress Summary	-124	262	131	53
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	9	9	9	9

Shear Stress Summary	6	6	12	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	124	263	133	58
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	2	2	-2	-2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	-3	-2	-8	-7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-88	88	-88	88
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-25	-25	25	25
Axial Bending (ML)	-135	135	135	-135
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-252	196	70	-23
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13

Shear Stress Summary	17	17	9	9
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	253	201	81	27
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-8	-8	-8	-8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	7	7	-7	-7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	2	2	-13	-13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-88	88	-88	88
Axial Membrane (Mc)	-25	-25	25	25
Axial Bending (Mc)	-147	147	147	-147
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-263	208	81	-34
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Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	13	13	13	13
Shear Stress Summary	9	9	17	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	266	208	101	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	89	90	89	90	89	90	89	90
Circ. Pl (SUS)	-6	-6	-10	-10	-20	-20	2	2
Circ. Q (SUS)	-87	87	23	-23	-193	193	39	-39
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	-20	-20	2	2	-1	-1	-15	-15
Long. Q (SUS)	-186	186	32	-32	-91	91	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	89.0	90.0	89.0	90.0	89.0	90.0	89.0	90.0
Pm+Pl (SUS)	83.1	84.1	79.1	80.1	69.2	70.1	91.1	92.1
Pm+Pl+Q (Total)	162.8	212.9	103.9	58.1	124.6	263.4	131.6	55.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	90.00	138.00	Passed
Pm+Pl (SUS)	92.06	207.00	Passed
Pm+Pl+Q (TOTAL)	263.38	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	-6	-6	-10	-10	-1	-1	-15	-15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-28	-28	22	22	-28	-28	22	22
Long. Q (SUS)	-223	223	47	-47	-235	235	59	-59



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	27.4	27.4	31.6	30.6	28.6	29.6	36.5	35.5
Pm+Pl+Q (Total)	251.0	201.6	80.5	28.6	263.8	208.5	100.2	44.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	118.00	Passed
Pm+Pl (SUS)	36.50	177.00	Passed
Pm+Pl+Q (TOTAL)	263.76	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N1-16"****From: 60**

Pressure for Reinforcement Calculations	P	6.152	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2899.9998	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4450.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	812.5000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		203.0500	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N1-16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.15 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.15)$
 $= 5.6012$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.15 \cdot 203.2 / (138 \cdot 1.0 + 0.4 \cdot 6.15)$
 $= 0.9043$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.0643 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 760.8000 mm.
 Parallel to Vessel Wall, opening length d 380.4000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 32.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	21.307	16.342	NA
Area in Shell	A1	28.145	16.768	NA
Area in Nozzle Wall	A2	7.862	7.758	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	42.528	42.528	NA
TOTAL AREA AVAILABLE	Atot	79.535	68.054	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.



The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)} \\
 &= 0.5(380.4 \cdot 8.592 \cdot 1 + 2 \cdot 13.0 \cdot 8.592 \cdot 1(1-1.0)) \\
 &= 16.342 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 380.4(1.0 \cdot 13.0 - 1.0 \cdot 8.592) - 2 \cdot 13.0 \\
 &\quad (1.0 \cdot 13.0 - 1.0 \cdot 8.592) \cdot (1 - 1.0) \\
 &= 16.768 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 32.5) \cdot (13.0 - 1.06) \cdot 1.0 \\
 &= 7.758 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned}
 &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\
 &= (760.8 - 406.4) \cdot 12.0 \cdot 1.0 \\
 &= 42.528 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0643 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.6012 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.6012 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.9395 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[11.331, \max(8.6012, 5.5)] \\
 &= 8.6012 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(4.0643, 8.6012) \\
 &= 8.6012 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.9,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	318.1 N./mm ²	Passed
Occasional	:	4.4,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	14.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\begin{aligned}
 &\text{Govrn. thk, } tg = 16.0, tr = 0.904, c = 3.0 \text{ mm.}, E^* = 1.0 \\
 &\text{Thickness Ratio} = tr \cdot E^* / (tg - c) = 0.07, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}
 \end{aligned}$$



Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.431$, Temp. Reduction = 44 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.431$, Temp. Reduction = 44 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.431$, Temp. Reduction = 44 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 5.601, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.431$, Temp. Reduction = 44 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.15/19.60 = 0.314$

Weld Size Calculations, Description: N1-16"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm.
 Pad Weld 6.0000 = $0.5 * T_{minPad}$ 8.4840 = $0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (16.3421 - 16.7679 + 2 \cdot 13.0 \cdot 1.0 \cdot$



$$(1.0 \cdot 13.0 - 8.592) \cdot 138 \\ = 1013.53 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr2}) \cdot \text{Sv} \\ = (7.7582 + 42.528 + 1. - 0.0 \cdot 1.0) \cdot 138 \\ = 72167.71 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr1})) \cdot \text{Sv} \\ = (7.7582 + 0.0 + 1. + (3.38)) \cdot 138 \\ = 17080.33 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr1})) \cdot \text{S} \\ = (7.7582 + 0.0 + 1. + 42.528 + (3.38)) \cdot 138 \\ = 76923.91 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw} \\ = (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 138 \\ = 44016. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot \text{DP} \cdot \text{WP} \cdot 0.49 \cdot \text{SEW} \\ = (3.1416/2.0) \cdot 812.5 \cdot 12.0 \cdot 0.49 \cdot 138 \\ = 105600. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo})/4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn} \\ = (3.1416 \cdot 196.7) \cdot (16.0 - 3.0) \cdot 0.7 \cdot 138 \\ = 79129. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgpn} \cdot 0.74 \cdot \text{Seg} \\ = (3.1416/2) \cdot 406.4 \cdot 12.0 \cdot 0.74 \cdot 138 \\ = 79768. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wgnvi} \cdot 0.74 \cdot \text{Sng} \\ = (3.1416/2.0) \cdot 406.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138 \\ = 66473. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (105600 + 79129) = 184729 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (44016 + 79768 + 66473 + 0) = 190258 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (105600 + 66473 + 0) = 172073 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 184729 Kgf, must exceed W = 1013 Kgf	or W1 = 72167 Kgf
Path 2-2 = 190257 Kgf, must exceed W = 1013 Kgf	or W2 = 17080 Kgf
Path 3-3 = 172073 Kgf, must exceed W = 1013 Kgf	or W3 = 76923 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 16.6267 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.6267 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.098 %

Input Echo, WRC297 Item 1, Description: N1-16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2500.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	16.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	812.500	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6.15	bars
Radial Load	P	3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{mean}(t - ca)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1265.5(16.0 - 3.0)}$$

$$= 829.669 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2531.000 mm.
Nozzle Outside Diameter	(d) =	406.400 mm.



Vessel Thickness used (T) = 25.000 mm.
 Nozzle Thickness used (t) = 13.000 mm.
 T / t = 1.923
 d / t = 31.262

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.616$$

Nr/P = 0.086
 Mr/P = 0.101
 M0/P = 0.056
 N0/P = 0.157
 MrD/Mc = 0.200
 NrDL/Mc = 0.112
 M0d/Mc = 0.128
 N0DL/Mc = 0.160
 MrD/Ml = 0.122
 NrDL/Ml = 0.077
 M0D/Ml = 0.069
 N0DL/Ml = 0.277

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-16	16	-16	16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	-53	51	67	27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-4	-4	-4	-4
Inplane Bending (P)	-29	29	-29	29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-89	98	53	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	90	98	68	46

Vessel Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-16	16	-16	16
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-67	67	67	-67
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	-90	77	73	-29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-4	-4	-4	-4
Inplane Bending (P)	-29	29	-29	29
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-105	105	105	-105
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31
Inplane Stress Summary	-118	151	112	-39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	118	151	113	42

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11
Hoop Stress Summary	-21	-20	27	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-61	61	-61	61



Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-14	-14	14	14
Axial Bending (ML)	-163	163	163	-163
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-236	214	119	-84
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	11	11	7	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	237	236	120	112

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11

Hoop Stress Summary	-11	-10	17	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-61	61	-61	61
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-281	281	281	-281
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-354	331	237	-201
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	7	7	11	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	354	342	238	220



Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1259.5(16.0 - 3.0)}$$

$$= 828.665 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2519.000 mm.

Nozzle Outside Diameter (d) = 812.500 mm.

Vessel Thickness used (T) = 13.000 mm.

Nozzle Thickness used (t) = 216.050 mm.

T / t = 0.060

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.490$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.049

Mr/P = 0.076

M0/P = 0.021

N0/P = 0.025

MrD/Mc = 0.245

NrDL/Mc = 0.105

M0d/Mc = 0.073

N0DL/Mc = 0.049

MrD/Ml = 0.053

NrDL/Ml = 0.056

M0D/Ml = 0.016

N0DL/Ml = 0.027

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	58	59	58	59
Outplane Stress Summary	11	88	52	65

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-80	80	-80	80
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	-120	144	1	58

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	133	144	52	66
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	-76	110	82	-16
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-80	80	-80	80
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-239	239	239	-239
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	59	58	59

Inplane Stress Summary	-287	354	226	-91
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	287	354	226	91
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	-31	-31	17	17	-13	-13	5	5
Circ. Q (SUS)	-52	52	20	-20	-134	134	76	-76
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-10	-10	2	2	-21	-21	7	7
Long. Q (SUS)	-93	93	35	-35	-83	83	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	6.3	5.4	47.0	48.0	23.1	24.1	35.1	36.1
Pm+Pl+Q (Total)	88.7	98.5	67.6	46.4	117.3	152.1	111.6	41.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+Pl (SUS)	48.03	207.00	Passed
Pm+Pl+Q (TOTAL)	152.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	59	58	59	58	59	58	59
Circ. Pl (SUS)	-8	-8	0	0	-25	-25	9	9
Circ. Q (SUS)	-37	37	-7	7	-319	319	159	-159
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-17	-17	1	1	-12	-12	4	4
Long. Q (SUS)	-131	131	-29	29	-93	93	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	59.0	58.0	59.0	58.0	59.0	58.0	59.0
Pm+Pl (SUS)	50.0	51.0	58.0	59.0	33.1	34.1	67.0	68.0
Pm+Pl+Q (Total)	132.1	143.2	51.0	66.1	286.0	353.0	226.1	91.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.00	138.00	Passed
Pm+Pl (SUS)	68.03	207.00	Passed
Pm+Pl+Q (TOTAL)	353.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-31	-31	17	17	-21	-21	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-15	-15	13	13	-15	-15	13	13
Long. Q (SUS)	-224	224	102	-102	-342	342	220	-220
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	21.4	20.4	27.4	28.4	12.6	12.0	19.6	20.0
Pm+Pl+Q (Total)	234.6	235.0	120.5	112.9	352.1	342.3	238.5	221.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	28.39	207.00	Passed
Pm+Pl+Q (TOTAL)	352.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N6-6"****From: 60**

Pressure for Reinforcement Calculations	P	6.086	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2899.9998	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5150.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	204.7870	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		75.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N6-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 6.625 in.
 Actual Thickness Used in Calculation 0.628 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.09 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.09)$
 $= 5.5412$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.09 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 6.09)$
 $= 0.4331$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5860 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 284.7198 mm.
 Parallel to Vessel Wall, opening length d 142.3599 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 22.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	8.097	6.277	NA
Area in Shell	A1	4.794	0.565	NA
Area in Nozzle Wall	A2	4.819	4.760	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	13.973	13.973	NA
TOTAL AREA AVAILABLE	Atot	24.442	20.154	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 169.0773 12.0000 mm.
 Based on given Pad Diameter: 318.2750 0.0827 mm.
 Based on Shell or Nozzle Thickness: 169.0773 12.0000 mm.



Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(142.3599 \cdot 8.592 \cdot 1 + 2 \cdot 12.9576 \cdot 8.592 \cdot 1(1 - 0.86))$$

$$= 6.277 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 142.36(1.0 \cdot 9.0 - 1.0 \cdot 8.592) - 2 \cdot 12.958$$

$$(1.0 \cdot 9.0 - 1.0 \cdot 8.592) \cdot (1 - 0.855)$$

$$= 0.565 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.5) \cdot (12.96 - 0.59) \cdot 0.8551$$

$$= 4.760 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (284.7198 - 168.275) \cdot 12.0 \cdot 1.0$$

$$= 13.973 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5860 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.5412 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.5412 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.9395 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(8.5412, 5.5)]$$

$$= 8.5412 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.586, 8.5412)$$

$$= 8.5412 \text{ mm.}$$

Available Nozzle Neck Thickness = 15.9576 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	272.2 N./mm ²	Passed
Occasional	:	1.5,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	14.6,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } tg = 15.958, tr = 0.433, c = 3.0 \text{ mm.}, E* = 1.0$$

$$\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.033, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A

6 °C



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.616$, Temp. Reduction = 21 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.616$, Temp. Reduction = 21 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 5.541, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.616$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.541, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.616$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.09/51.00 = 0.119$

Weld Size Calculations, Description: N6-6"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	4.5000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$
 $= \max(0, (6.2772 - 0.5654 + 2 \cdot 12.9576 \cdot 0.8551 \cdot (1.0 \cdot 9.0 - 8.592))138)$
 $= 8164.49 \text{ Kg}$



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr2}) \cdot S_v \\
 &= (4.7604 + 13.9734 + 0.8551 - 0.0 \cdot 0.86) \cdot 138 \\
 &= 27564.55 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v \\
 &= (4.7604 + 0.0 + 0.8551 + (1.9943)) \cdot 138 \\
 &= 10708.16 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S \\
 &= (4.7604 + 0.0 + 0.8551 + 13.9734 + (1.9943)) \cdot 138 \\
 &= 30370.89 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 15584. \text{ Kgf}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\
 &= (3.1416/2.0) \cdot 318.275 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 34472. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 77.6587) \cdot (15.9576 - 3.0) \cdot 0.7 \cdot 118 \\
 &= 26626. \text{ Kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\
 &= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 138 \\
 &= 33029. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 138 \\
 &= 24772. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (34472 + 26626) = 61098 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (15584 + 33029 + 24772 + 0) = 73385 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\
 &= (34472 + 24772 + 0) = 59243 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 61097	Kgf, must exceed W = 8164 Kgf	or W1 = 27564 Kgf
Path 2-2 = 73384	Kgf, must exceed W = 8164 Kgf	or W2 = 10708 Kgf
Path 3-3 = 59243	Kgf, must exceed W = 8164 Kgf	or W3 = 30370 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 8 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 2.8348 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.6219 mm.

Input Echo, WRC297 Item 1, Description: N6-6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2500.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	168.275	mm.
Nozzle Thickness	15.9576	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6.09	bars
Radial Load	P	1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1263.5(12.0 - 3.0))} \\
 &= 520.178 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2527.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	21.000	mm.
Nozzle Thickness used	(t) =	12.958	mm.
T / t	=	1.621	
d / t	=	12.987	



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.730$$

Nr/P	=	0.147
Mr/P	=	0.182
M0/P	=	0.086
N0/P	=	0.160
MrD/Mc	=	0.247
NrDL/Mc	=	0.165
M0d/Mc	=	0.143
N0DL/Mc	=	-0.075
MrD/M1	=	0.205
NrDL/M1	=	0.145
M0D/M1	=	0.117
N0DL/M1	=	0.102

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	36	36	36	36
Outplane Stress Summary	-15	71	52	20

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-27	27	-27	27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-71	87	44	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	72	89	54	24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-4	-4	-4	-4



Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-36	36	36	-36
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18

Outplane Stress Summary	-32	66	34	-12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-27	27	-27	27
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-62	62	62	-62
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	36	36	36

Inplane Stress Summary	-65	116	74	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	65	116	75	21
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	-4	-3	3	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-43	43	-43	43
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-88	88	88	-88
Axial Pressure Stress	1	1	1	1



Axial Stress Summary	-145	117	58	-30
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	146	123	59	37
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	4	3	4

Hoop Stress Summary	2	3	-3	-2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-43	43	-43	43
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-109	109	109	-109
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-166	139	80	-52
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	170	139	86	54
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:



$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1257.5(12.0 - 3.0))} \\
 &= 519.341 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2515.000 mm.
Nozzle Outside Diameter	(d) =	318.275 mm.
Vessel Thickness used	(T) =	9.000 mm.
Nozzle Thickness used	(t) =	87.958 mm.
T / t	=	0.102
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.115$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.082
Mr/P	=	0.141
M0/P	=	0.042
N0/P	=	0.052
MrD/Mc	=	0.288
NrDL/Mc	=	0.112
M0d/Mc	=	0.088
N0DL/Mc	=	0.061
MrD/M1	=	0.106
NrDL/M1	=	0.109
M0D/M1	=	0.032
N0DL/M1	=	0.070

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-34	34	-34	34
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	84	85	84	85
Outplane Stress Summary	10	127	74	97

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-117	117	-117	117
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	42	42	42	42
Inplane Stress Summary	-177	212	4	84

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3



Shear Stress Summary 4 4 1 1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 188 212 74 97

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-34	34	-34	34
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-64	64	64	-64
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary -71 126 72 12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	-11	-11	-11	-11
Inplane Bending (P)	-117	117	-117	117
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-211	211	211	-211
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	85	84	85

Inplane Stress Summary -269 388 180 -6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 1 1 4 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 269 388 180 21

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	36	36	36	36	36	36	36
Circ. Pl (SUS)	-8	-8	0	0	-9	-9	3	3
Circ. Q (SUS)	-42	42	16	-16	-89	89	35	-35



Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-9	-9	3	3	-1	-1	-7	-7
Long. Q (SUS)	-79	79	25	-25	-49	49	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
Pm+Pl (SUS)	28.1	28.1	36.1	36.1	27.1	27.1	39.0	39.0
Pm+Pl+Q (Total)	70.4	89.3	53.2	24.7	62.3	116.2	74.6	18.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	39.04	207.00	Passed
Pm+Pl+Q (TOTAL)	116.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	85	84	85	84	85	84	85
Circ. Pl (SUS)	-15	-15	1	1	-24	-24	2	2
Circ. Q (SUS)	-57	57	-11	11	-328	328	94	-94
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-24	-24	2	2	-14	-14	0	0
Long. Q (SUS)	-194	194	-40	40	-98	98	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	84.0	85.0	84.0	85.0	84.0	85.0	84.0	85.0
Pm+Pl (SUS)	69.0	70.0	85.0	86.0	60.0	61.0	86.0	87.0
Pm+Pl+Q (Total)	188.2	212.2	74.1	97.3	268.0	389.0	180.1	20.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	85.00	138.00	Passed
Pm+Pl (SUS)	87.02	207.00	Passed
Pm+Pl+Q (TOTAL)	389.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:
Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	4	3	4	3	4	3	4
Circ. Pl (SUS)	-8	-8	0	0	-1	-1	-7	-7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-14	-14	12	12	-14	-14	12	12
Long. Q (SUS)	-131	131	45	-45	-152	152	66	-66
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	3.0	4.0	3.0	4.0	3.0	4.0	3.0	4.0
Pm+Pl (SUS)	13.5	13.4	13.4	13.4	15.5	16.5	17.5	16.5
Pm+Pl+Q (Total)	144.6	123.3	58.5	37.4	167.3	139.2	84.9	54.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	118.00	Passed
Pm+Pl (SUS)	17.46	177.00	Passed
Pm+Pl+Q (TOTAL)	167.30	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N4-3"****From: 60**

Pressure for Reinforcement Calculations	P	6.096	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2500.00	mm.
Design Length of Section	L	2899.9998	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5050.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-106 B	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N4-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.810	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.1 \cdot 1253.0 / (138 \cdot 1.0 - 0.6 \cdot 6.1)$
 $= 5.5497$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.1 \cdot 41.1 / (118 \cdot 1.0 - 0.6 \cdot 6.1)$
 $= 0.2130$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000	mm.
Parallel to Vessel Wall, opening length	d	82.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N4-3".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4707	mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.5497	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.5497	mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.9395	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000	mm.
Wall Thickness per table UG-45	tb3	= 8.7300	mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.73, \max(8.5497, 5.5)]$
 $= 8.5497$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(3.4707, 8.5497)$
 $= 8.5497$ mm.

Available Nozzle Neck Thickness = 20.5740 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.2,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	282.8 N./mm ²	Passed
Occasional	:	0.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	9.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.55, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.617$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 6.10/51.00 = 0.120

Weld Size Calculations, Description: N4-3"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.3778 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 213.4029 mm.

Input Echo, WRC297 Item 1, Description: N4-3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2500.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	76.200	mm.
Nozzle Thickness	20.5740	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	6.10	bars
Radial Load	P	611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf
Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.704) is < 10
 check the Limitations of WRC bulletin 297.



Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 117.348 + 2 \cdot 1.65 \cdot \sqrt{(1257.5(12.0 - 3.0))}$$

$$= 468.414 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2515.000 mm.
Nozzle Outside Diameter	(d) =	117.348 mm.
Vessel Thickness used	(T) =	9.000 mm.
Nozzle Thickness used	(t) =	17.574 mm.
T / t	=	0.512
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.780$$

Nr/P	=	0.118
Mr/P	=	0.258
M0/P	=	0.076
N0/P	=	0.147
MrD/Mc	=	0.309
NrDL/Mc	=	0.088
M0d/Mc	=	0.098
N0DL/Mc	=	0.022
MrD/M1	=	0.226
NrDL/M1	=	0.091
M0D/M1	=	0.074
N0DL/M1	=	0.098

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-33	33	-33	33
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-41	41	41	-41
Normal Pressure Stress	84	85	84	85
Outplane Stress Summary	-11	140	90	75

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-114	114	-114	114
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-127	127	127	-127
Inplane Pressure Stress	42	42	42	42
Inplane Stress Summary	-217	267	54	29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	6	6	6	6
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Shear Stress Summary	8	8	4	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	217	268	91	75
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-10	-10	-10	-10
Outplane Bending (P)	-33	33	-33	33
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-55	55	55	-55
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary	-59	118	55	11
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-114	114	-114	114
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-174	174	174	-174
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	85	84	85

Inplane Stress Summary	-221	357	144	24
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	222	357	144	29
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2
Hoop Stress Summary	-18	-18	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-23	23	-23	23
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-26	26	26	-26
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-57	42	10	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	58	60	11	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-10	-10	-10	-10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	2	1	2
Hoop Stress Summary	-11	-10	-7	-6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-23	23	-23	23
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-39	39	39	-39
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-70	54	23	-9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	70	65	31	12

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	85	84	85	84	85	84	85
Circ. Pl (SUS)	-19	-19	-1	-1	-16	-16	0	0
Circ. Q (SUS)	-74	74	8	-8	-288	288	60	-60
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-16	-16	0	0	-12	-12	-8	-8
Long. Q (SUS)	-241	241	13	-13	-88	88	22	-22
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	84.0	85.0	84.0	85.0	84.0	85.0	84.0	85.0
Pm+Pl (SUS)	65.1	66.1	83.1	84.1	68.1	69.1	84.1	85.1
Pm+Pl+Q (Total)	215.3	267.5	91.4	76.3	220.1	357.1	144.7	28.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	85.00	138.00	Passed
Pm+Pl (SUS)	85.08	207.00	Passed
Pm+Pl+Q (TOTAL)	357.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	2	1	2	1	2	1	2
Circ. Pl (SUS)	-19	-19	-1	-1	-12	-12	-8	-8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-8	-8	6	6	-8	-8	6	6
Long. Q (SUS)	-49	49	3	-3	-62	62	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1



Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
Pm+Pl (SUS)	18.1	17.1	6.3	6.2	11.3	10.4	13.2	12.2
Pm+Pl+Q (Total)	57.4	58.5	9.8	4.5	70.1	64.1	30.1	12.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	118.00	Passed
Pm+Pl (SUS)	18.10	177.00	Passed
Pm+Pl+Q (TOTAL)	70.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N3-20"****From: 70**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	622.0000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		57.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N3-20"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 20.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.05 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.05)$
 $= 4.9340$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.05 \cdot 254.0 / (138 \cdot 1.0 + 0.4 \cdot 6.05)$
 $= 1.1114$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.9797 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 980.0001 mm.
 Parallel to Vessel Wall, opening length d 490.0000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	24.177	15.986	NA
Area in Shell	A1	29.723	21.928	NA
Area in Nozzle Wall	A2	4.339	4.411	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.640	1.640	NA
Area in Element	A5	13.680	13.680	NA
TOTAL AREA AVAILABLE	Atot	49.382	41.660	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (490.0 \cdot 4.934 \cdot 1.0 + 2 \cdot 9.0 \cdot 4.934 \cdot 1.0 \cdot (1 - 1.0))$
 $= 24.177$ cm²

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 490.0(1.0 \cdot 11.0 - 1.0 \cdot 4.934) - 2 \cdot 9.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 4.934) \cdot (1 - 1.0)$$

$$= 29.723 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 27.5) \cdot (9.0 - 1.11) \cdot 1.0$$

$$= 4.339 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 8.0^2 \cdot 1.0$$

$$= 1.640 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (622.0 - 508.0) \cdot 12.0 \cdot 1.0$$

$$= 13.680 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1114 mm.
Wall Thickness per UG16(b),	tr16b = 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 8.4773 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 8.4773 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.9362 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(8.4773, 5.5)]$$

$$= 8.4773 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.1114, 8.4773)$$

$$= 8.4773 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	34.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	311.0 N./mm ²	Passed
Occasional	:	8.1,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	19.8,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.0, tr = 1.111, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.123, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, tr = 1.111, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.123, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, tr = 1.111, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.123$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.449$, Temp. Reduction = 40 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 1.111, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.123$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.05/19.60 = 0.309$

Weld Size Calculations, Description: N3-20"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.0000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	5.5000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2*tn*fr1*(E1*t-tr))*Sv)$
 $= \max(0, (24.1768 - 29.7232 + 2 * 9.0 * 1.0 * (1.0 * 11.0 - 4.934)) * 138)$
 $= \max(0, -6268.30) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv$
 $= (4.3387 + 13.68 + 1.64 - 0.0 * 1.0) * 138$
 $= 27662.90 \text{ Kg}$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (4.3387 + 0.0 + 1. + (1.98)) \cdot 138$$

$$= 10298.58 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (4.3387 + 0.0 + 1.64 + 13.68 + (1.98)) \cdot 138$$

$$= 30449.08 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 55020. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 622.0 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 53894. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 249.5) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 69487. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 508.0 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 99710. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (9.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 74783. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (53894 + 69487) = 123381 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (55020 + 99710 + 74783 + 0) = 229513 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (53894 + 74783 + 0) = 128676 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 123380 Kg, must exceed W = 0 Kg	or W1 = 27662 Kg
Path 2-2 = 229512 Kg, must exceed W = 0 Kg	or W2 = 10298 Kg
Path 3-3 = 128676 Kg, must exceed W = 0 Kg	or W3 = 30449 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 14.3111 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 178.3111 mm.

Percent Elongation Calculations:



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.419 %

Input Echo, WRC107/537 Item 1, Description: N3-20" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	4500.000	mm.
Vessel Thickness	Tv	14.000	mm.
Design Temperature	T1	180.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm.
Nozzle Diameter	Dn	508.000	mm.
Nozzle Thickness	Tn	12.000	mm.
Nozzle Material		SA-516 70	
Nozzle UNS Number		K02700	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	12.000	mm.
Diameter of Reinforcing Pad	Dpad	622.000	mm.
Design Internal Pressure	Dp	6.049	bars
Include Pressure Thrust		No	

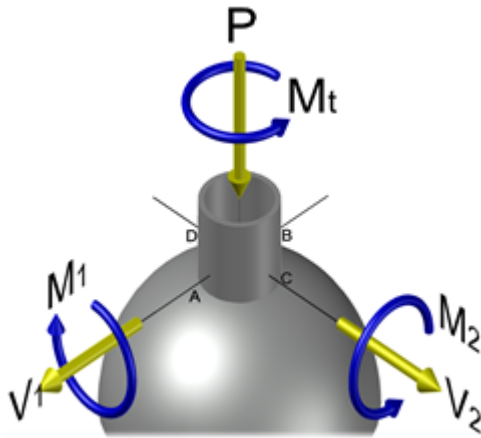
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3824.2	Kgf
Longitudinal Shear (SUS)	(V1) V1	2549.5	Kgf
Circumferential Shear (SUS)	(Vc) V2	2549.5	Kgf
Circumferential Moment (SUS)	(Mc) M1	3567.9	Kg-m.
Longitudinal Moment (SUS)	(M1) M2	3567.9	Kg-m.
Torsional Moment (SUS)	Mt	5097.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(2258.5 (14.0 - 3.0))}$$

$$= 1028.140 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-3824.2	Kgf
Circumferential Shear	(VC)	V2	2549.5	Kgf
Longitudinal Shear	(VL)	V1	2549.5	Kgf
Circumferential Moment	(MC)	M1	3567.9	Kg-m.
Longitudinal Moment	(ML)	M2	3567.9	Kg-m.
Torsional Moment		MT	5097.0	Kg-m.

Dimensionless Param: U = 1.11 TAU = 27.72 RHO = 4.00 (2.56)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 7	0.02129	(A,B,C,D)
$M(x) / P$	SP 7	0.00828	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.03252	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 7	0.01403	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.03252	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 7	0.01403	(A,B,C,D)
$N(y) * T / P$	SP 7	0.18929	(A,B,C,D)
$M(y) / P$	SP 7	0.02115	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 7	0.23004	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 7	0.04487	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 7	0.23004	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 7	0.04487	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Membr. P	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Rad.	Bend. P	3.5	-3.5	3.5	-3.5	3.5	-3.5	3.5	-3.5
Rad.	Membr. MC	0.0	0.0	0.0	0.0	-9.4	-9.4	9.4	9.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-24.4	24.4	24.4	-24.4
Rad.	Membr. ML	-9.4	-9.4	9.4	9.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-24.4	24.4	24.4	-24.4	0.0	0.0	0.0	0.0

Tot. Rad. Str.	-28.8	13.0	38.9	-17.0	-28.8	13.0	38.9	-17.0
Tang. Memb. P	13.4	13.4	13.4	13.4	13.4	13.4	13.4	13.4
Tang. Bend. P	9.0	-9.0	9.0	-9.0	9.0	-9.0	9.0	-9.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-66.7	-66.7	66.7	66.7
Tang. Bend. MC	0.0	0.0	0.0	0.0	-78.0	78.0	78.0	-78.0
Tang. Memb. ML	-66.7	-66.7	66.7	66.7	0.0	0.0	0.0	0.0
Tang. Bend. ML	-78.0	78.0	78.0	-78.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-122.3	15.8	167.1	-6.9	-122.3	15.8	167.1	-6.9
Shear VC	1.4	1.4	-1.4	-1.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Tot. Shear	6.7	6.7	4.0	4.0	4.0	4.0	6.7	6.7
Str. Int.	122.8	21.2	167.2	18.4	122.4	18.6	167.5	20.4

Unitless Prm: U = 1.97 TAU = 0.00 (34.06) RHO = 0.00 (1.22)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02529	(A,B,C,D)
M(x) / P	SR 2	0.01187	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.02123	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01389	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.02123	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01389	(A,B,C,D)
N(y) * T / P	SR 2	0.00757	(A,B,C,D)
M(y) / P	SR 2	0.00357	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00636	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00414	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00636	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00414	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	7.8	7.8	7.8	7.8	7.8	7.8	7.8	7.8
Rad.	Bend. P	22.1	-22.1	22.1	-22.1	22.1	-22.1	22.1	-22.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-38.9	-38.9	38.9	38.9
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-152.9	152.9	152.9	-152.9
Rad.	Memb. ML	-38.9	-38.9	38.9	38.9	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-152.9	152.9	152.9	-152.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-161.9	99.7	221.8	-128.2	-161.9	99.7	221.8	-128.2
Tang. Memb. P		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tang. Bend. P		6.6	-6.6	6.6	-6.6	6.6	-6.6	6.6	-6.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-11.7	-11.7	11.7	11.7
Tang. Bend. MC		0.0	0.0	0.0	0.0	-45.6	45.6	45.6	-45.6
Tang. Memb. ML		-11.7	-11.7	11.7	11.7	0.0	0.0	0.0	0.0
Tang. Bend. ML		-45.6	45.6	45.6	-45.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-48.3	29.6	66.2	-38.2	-48.3	29.6	66.2	-38.2
Shear VC		2.3	2.3	-2.3	-2.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Shear MT		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5



Tot. Shear	9.8	9.8	5.2	5.2	5.2	5.2	9.8	9.8
Str. Int.	162.8	101.0	221.9	128.5	162.2	100.1	222.4	129.2

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		29.6	29.6	29.6	29.6	29.6	29.6	29.6	29.6
Rad. Pl (SUS)		-7.9	-7.9	10.9	10.9	-7.9	-7.9	10.9	10.9
Rad. Q (SUS)		-20.9	20.9	27.9	-27.9	-20.9	20.9	27.9	-27.9
Long. Pm (SUS)		29.6	29.6	29.6	29.6	29.6	29.6	29.6	29.6
Long. Pl (SUS)		-53.2	-53.2	80.1	80.1	-53.2	-53.2	80.1	80.1
Long. Q (SUS)		-69.0	69.0	87.0	-87.0	-69.0	69.0	87.0	-87.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.4	1.4	-1.4	-1.4	-1.4	-1.4	1.4	1.4
Shear Q (SUS)		5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Pm (SUS)		29.6	29.6	29.6	29.6	29.6	29.6	29.6	29.6
Pm+Pl (SUS)		45.4	45.4	109.7	109.7	45.4	45.4	109.7	109.7
Pm+Pl+Q (Total)		94.4	50.9	196.9	24.1	93.8	48.2	197.1	26.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.63	138.00	Passed
Pm+Pl (SUS)	109.74	207.00	Passed
Pm+Pl+Q (TOTAL)	197.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		61.9	61.9	61.9	61.9	61.9	61.9	61.9	61.9
Rad. Pl (SUS)		-31.1	-31.1	46.8	46.8	-31.1	-31.1	46.8	46.8
Rad. Q (SUS)		-130.8	130.8	175.0	-175.0	-130.8	130.8	175.0	-175.0
Long. Pm (SUS)		61.9	61.9	61.9	61.9	61.9	61.9	61.9	61.9
Long. Pl (SUS)		-9.3	-9.3	14.0	14.0	-9.3	-9.3	14.0	14.0
Long. Q (SUS)		-38.9	38.9	52.2	-52.2	-38.9	38.9	52.2	-52.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.3	2.3	-2.3	-2.3	-2.3	-2.3	2.3	2.3
Shear Q (SUS)		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Pm (SUS)		61.9	61.9	61.9	61.9	61.9	61.9	61.9	61.9
Pm+Pl (SUS)		52.9	52.9	108.9	108.9	52.9	52.9	108.9	108.9
Pm+Pl+Q (Total)		115.3	163.0	283.9	90.6	114.1	162.0	284.3	92.1

**Vessel Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.95	138.00	Passed
Pm+Pl (SUS)	108.90	207.00	Passed
Pm+Pl+Q (TOTAL)	284.33	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: K1A-2"****From: 70**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	180	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2500.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	9.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight



of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.05 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.05)$
 $= 4.9340$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 0.1248$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	117.7549	mm.
Parallel to Vessel Wall, opening length	d	58.8775	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govern. thk, tg = 10.0, tr = 4.934, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.705$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.05 / 51.00 = 0.119$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot Wo$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

**Messages regarding the Nozzle Input Data:**

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.05 \cdot 2506.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 6.05)$
 $= 4.9340 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 0.1248 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3924 mm.
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 8.4773 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.4773 mm.
Wall Thickness, shell/head, external pressure	trb2	= 3.9362 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm.
Wall Thickness per table UG-45	tb3	= 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(8.4773, 5.5)]$
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3924, 7.8)$
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 12.0636 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.8097 mm.

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